

LinguaQuanta: Towards a Quantum Transpiler

Scott Wesley
Dalhousie University

A dark blue diagonal gradient bar that starts from the bottom left and extends towards the top right, covering the lower half of the slide.

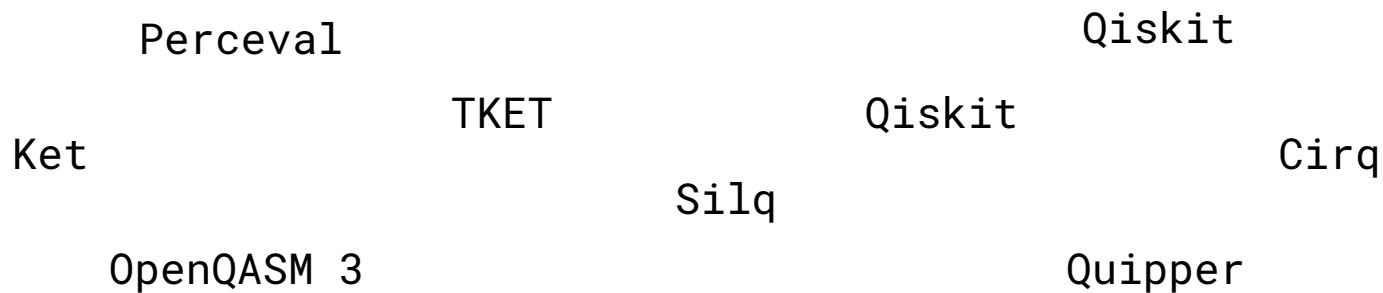
Outline of Talk

1. **Motivations and Background**
2. **Correctness:** Categorical Specifications
3. **Translations:** Some Decompositions
4. **Challenges:** Ancilla Management

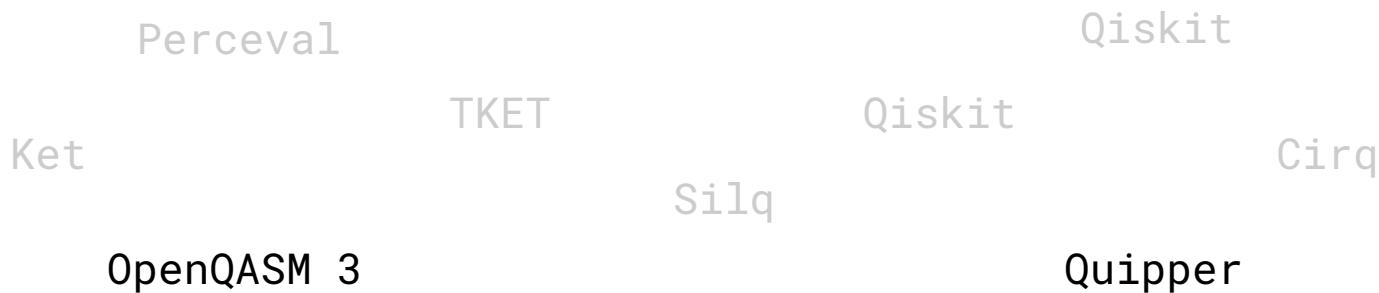
Objectives and Overview

A Story of Interoperability

A Story of Interoperability



A Story of Interoperability



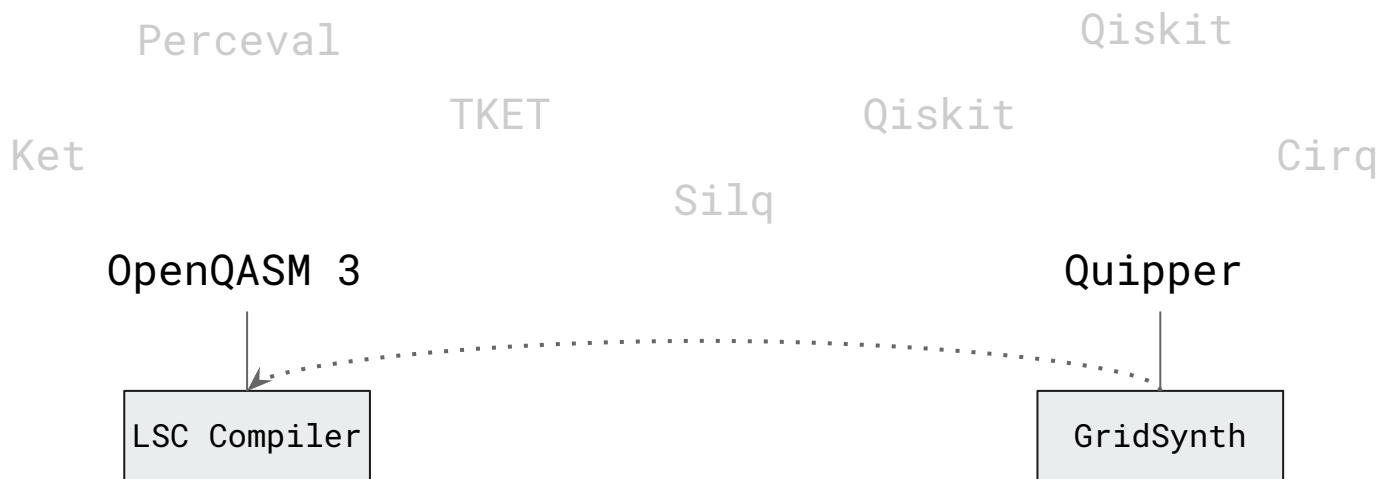
A Story of Interoperability



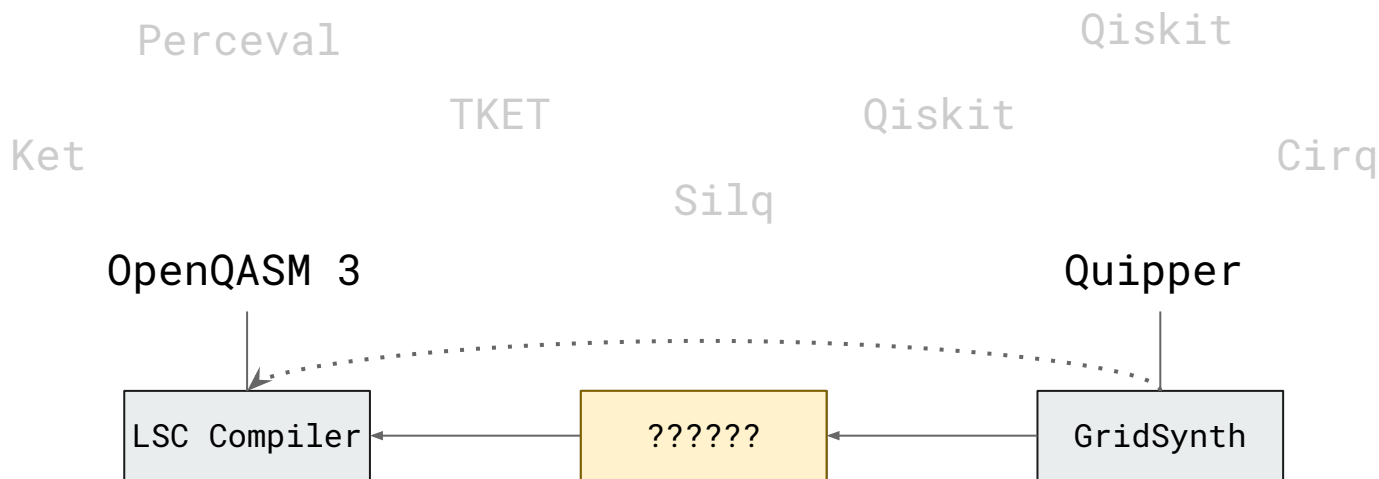
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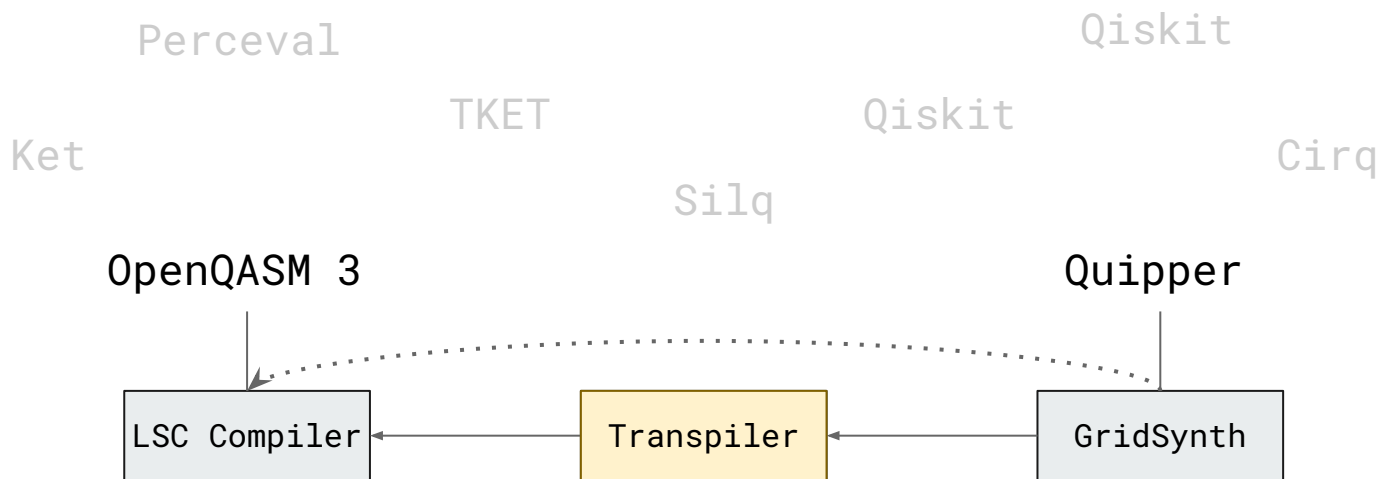
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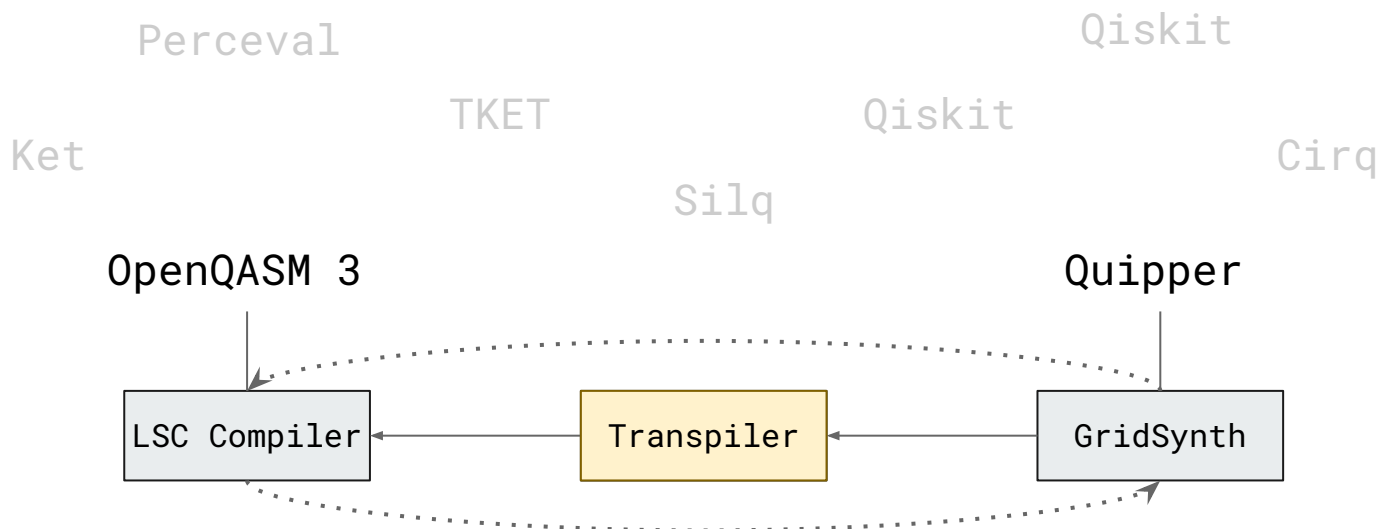
A Story of Interoperability



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A Story of Interoperability



Categorical Specification

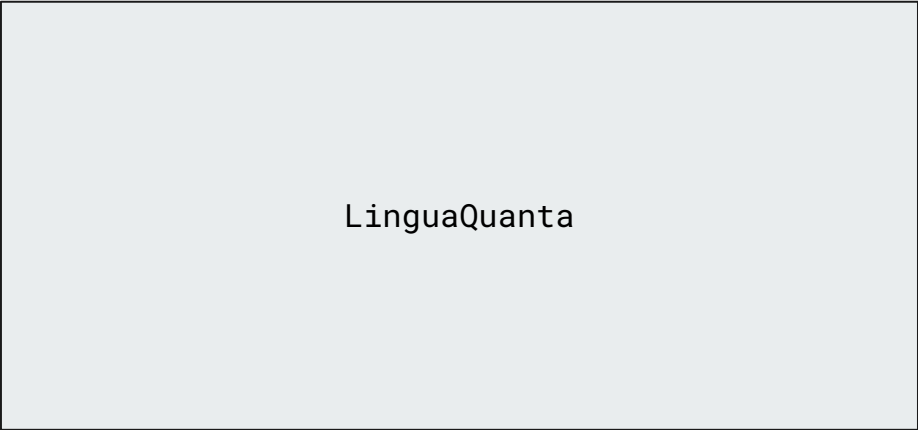
How Should an Ideal Transpiler Behave?



The UNIX Design Philosophy and Pipelines

Implications of the UNIX philosophy:

- 1.
- 2.
- 3.



LinguaQuanta

The UNIX Design Philosophy and Pipelines

Implications of the UNIX philosophy:

1. Write programs that do one thing and do it well.
- 2.
- 3.

LinguaQuanta

ElimInvs

ElimCtrls

ElimPows

ElimFuns

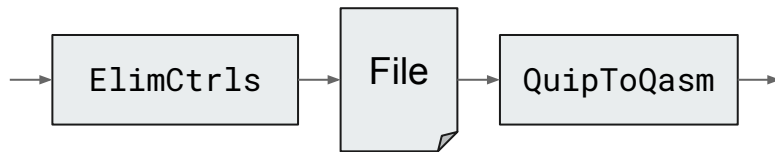
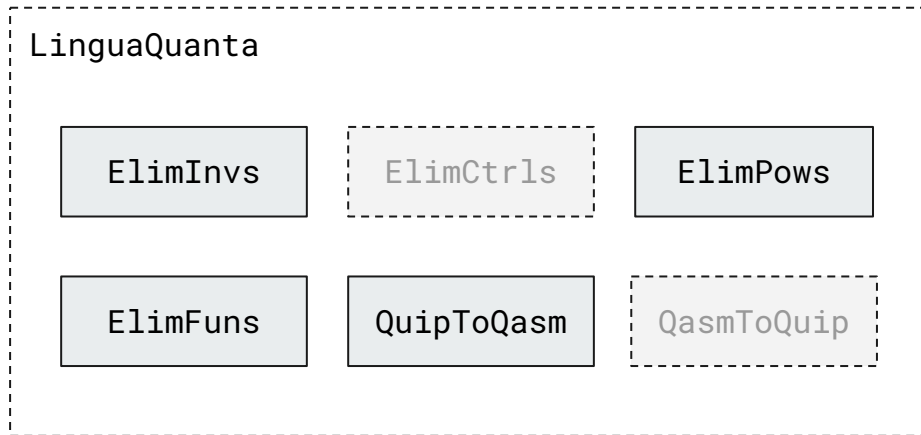
QuipToQasm

QasmToQuip

The UNIX Design Philosophy and Pipelines

Implications of the UNIX philosophy:

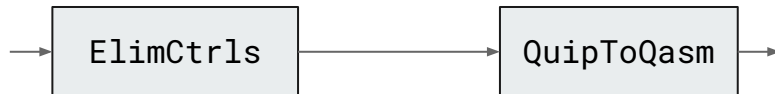
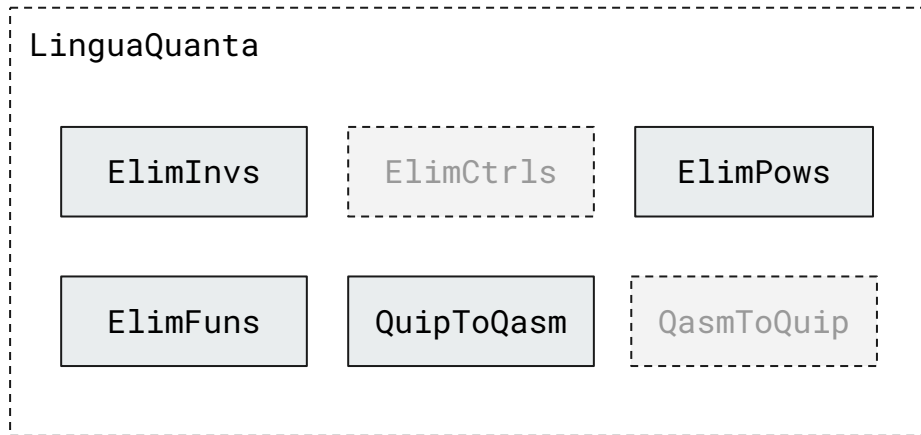
1. Write programs that do one thing and do it well.
2. Write programs that work well together.
- 3.



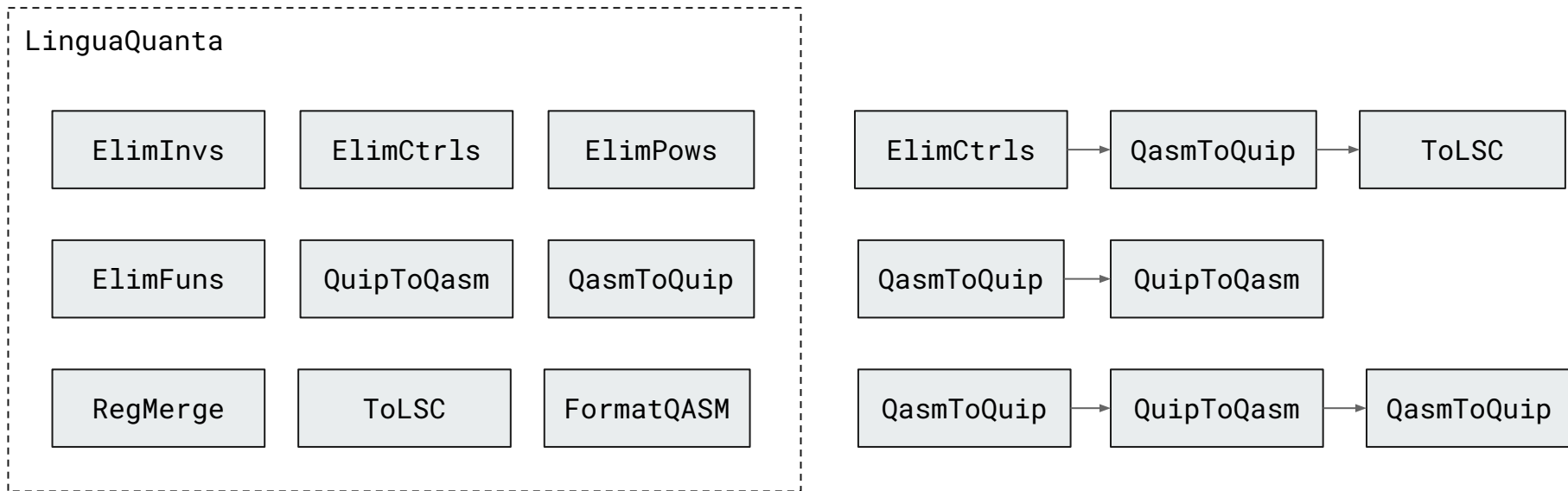
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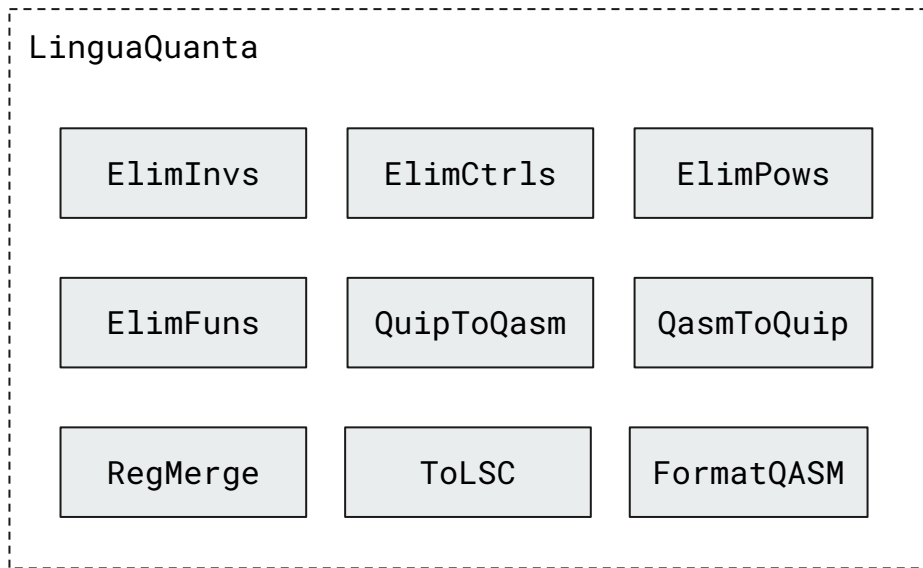
1. Write programs that do one thing and do it well.
2. Write programs that work well together.
3. Write programs that handle text streams.



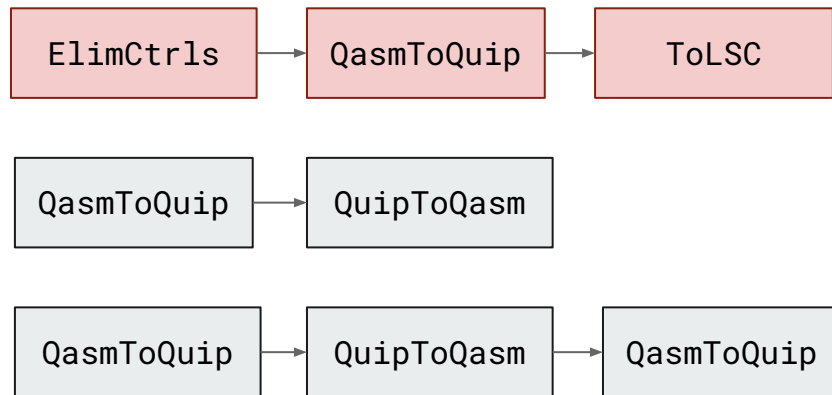
Problem: Many Potential Pipelines



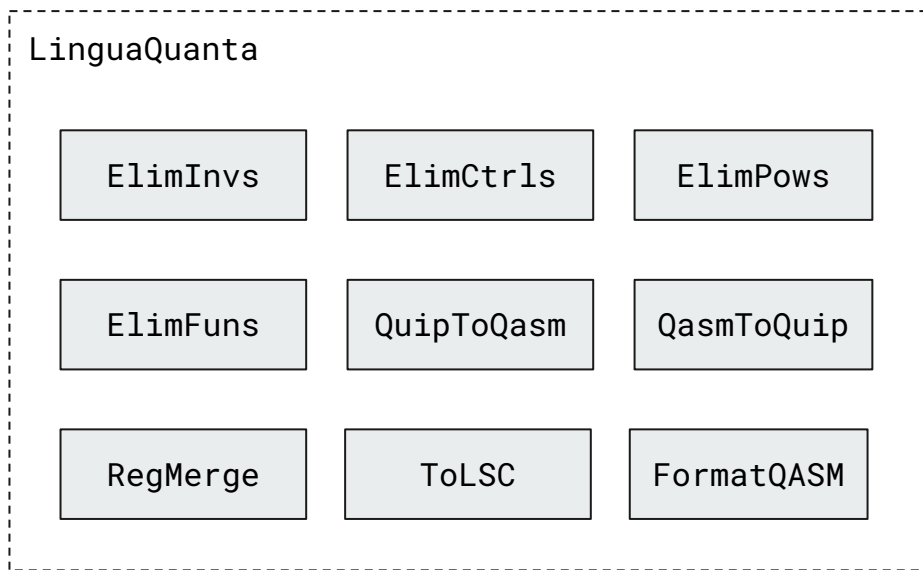
Problem: Many Potential Pipelines



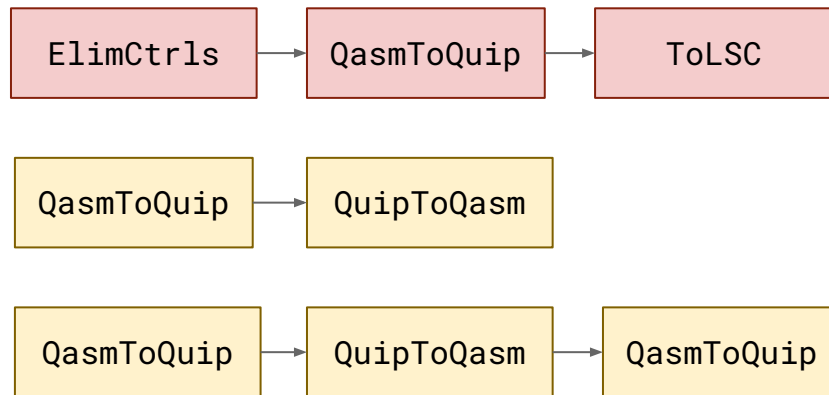
The output file types do not align with the input file types.



Problem: Many Potential Pipelines

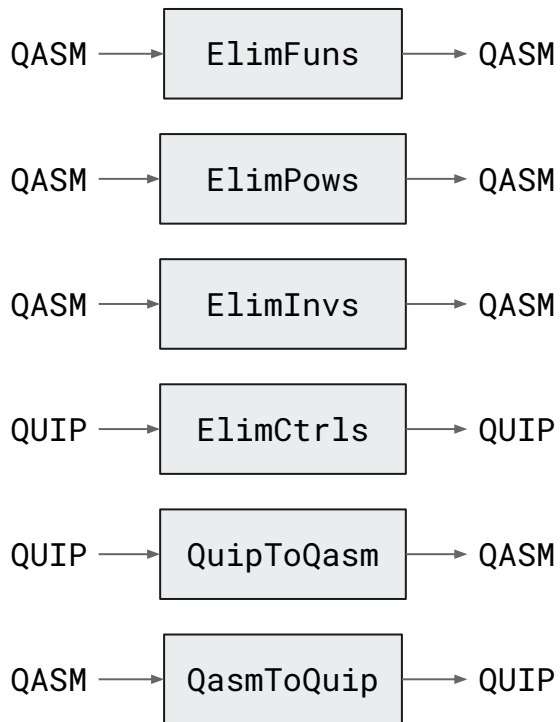


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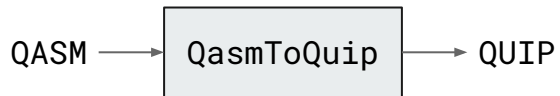
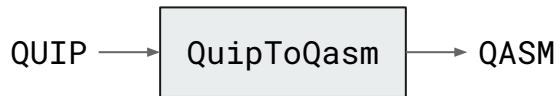
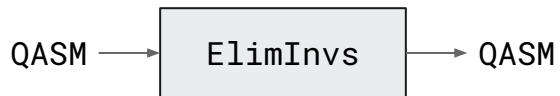
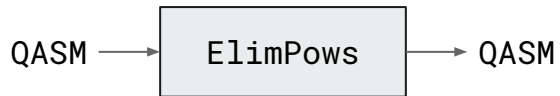
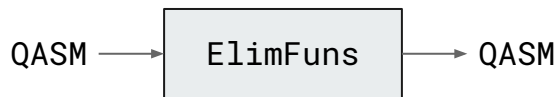


What should we expect from a round translation in LinguaQuanta?

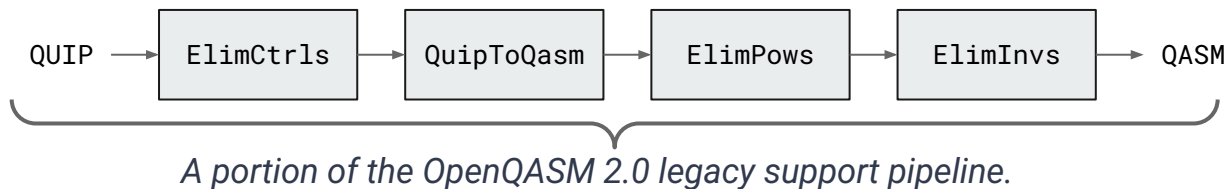
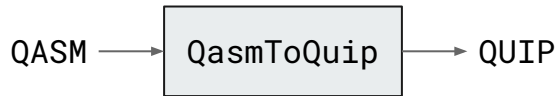
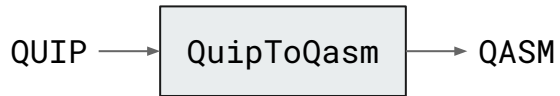
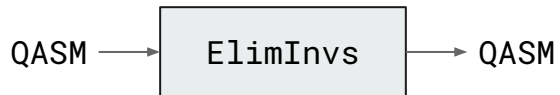
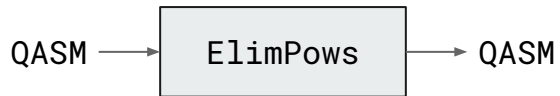
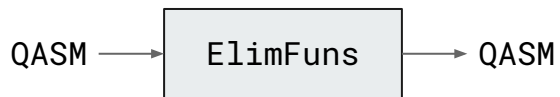
Solution: Study the Semantics of Pipelines



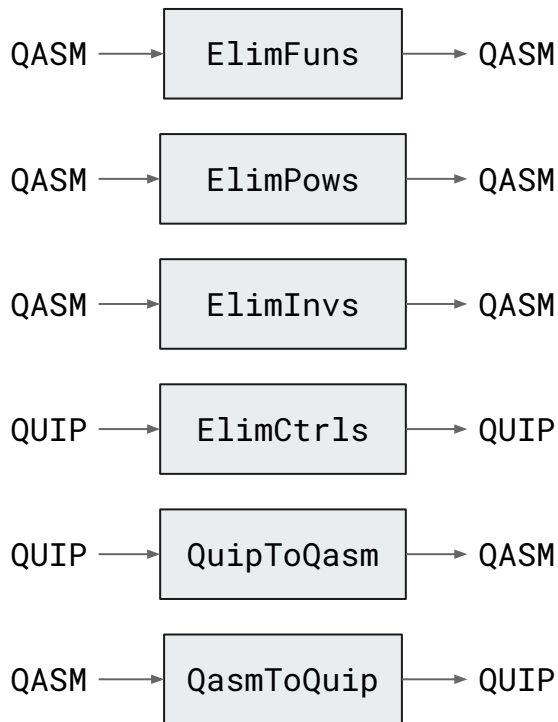
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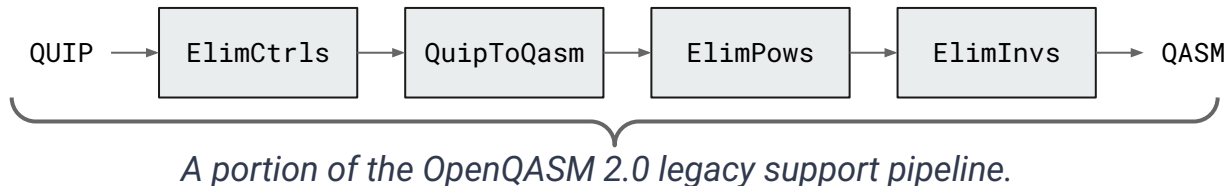


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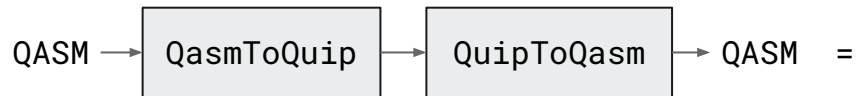
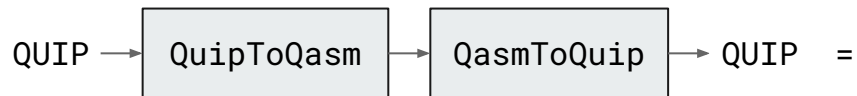
Now our pipelines are **type safe** in some very coarse grained sense. However, we would like to have a more fine-grained **notion of correctness**. Ideally, we would be able to **validate these specifications algorithmically**.

The rest of this section will refine these specifications to a satisfactory level, and then suggest a **direction for future work**.



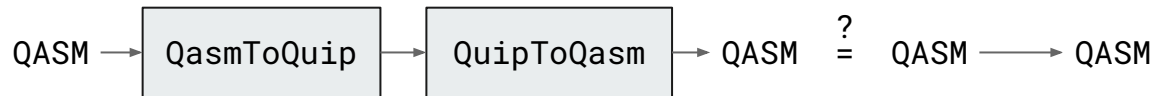
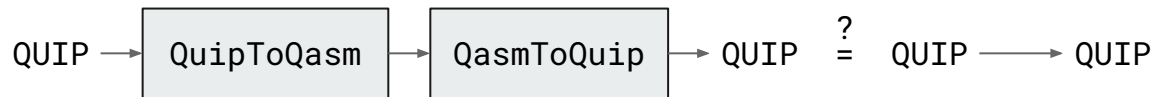
Step 1: Round Translations (Idempotents)

Q. What should happen to a round trip translation?



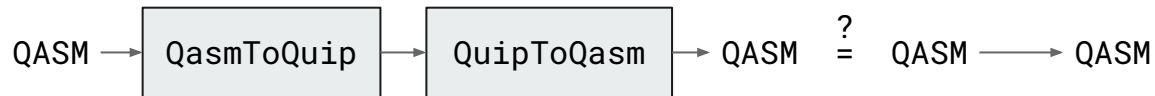
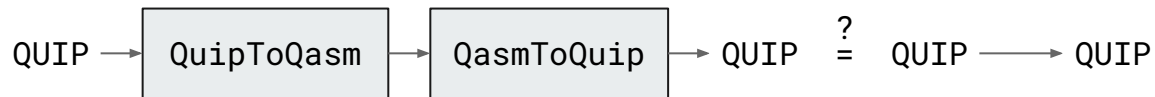
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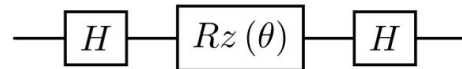
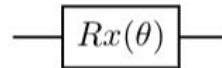


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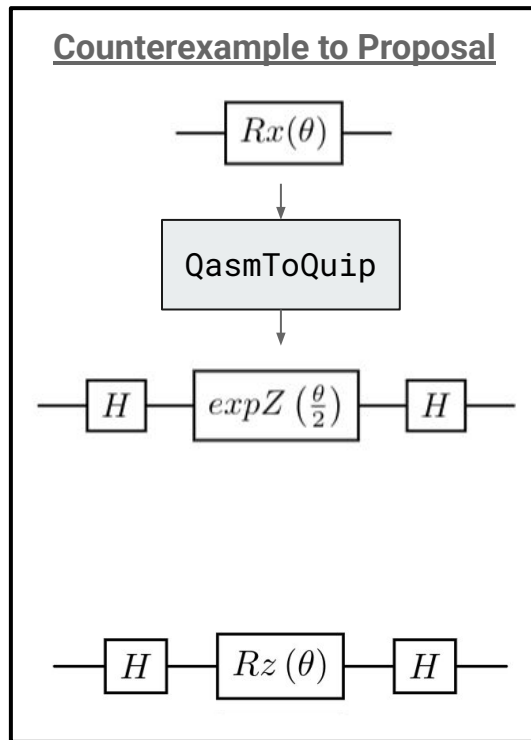
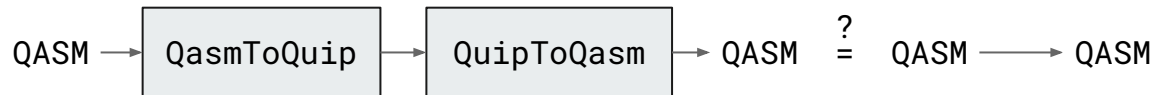
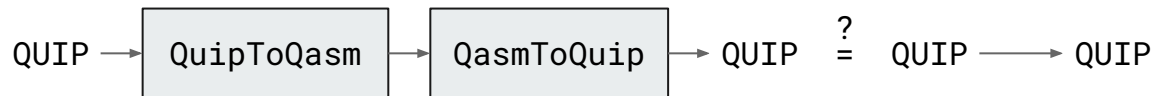


Counterexample to Proposal



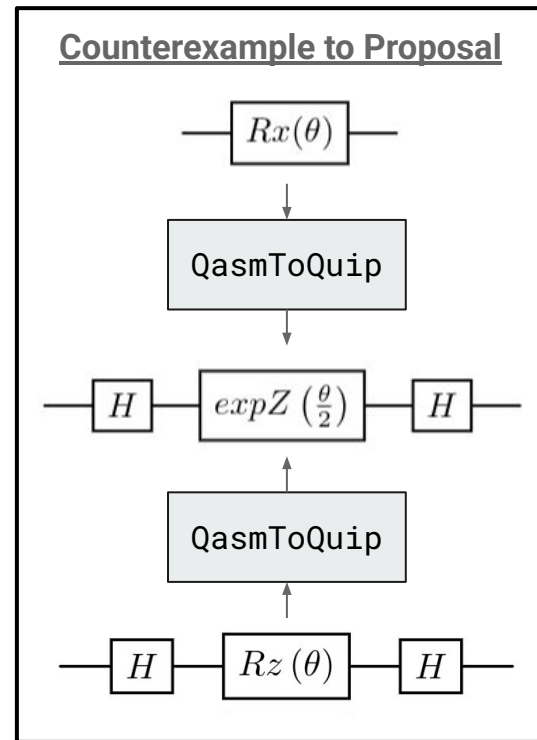
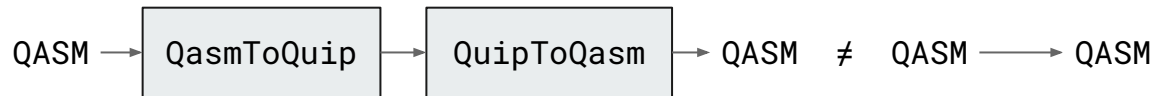
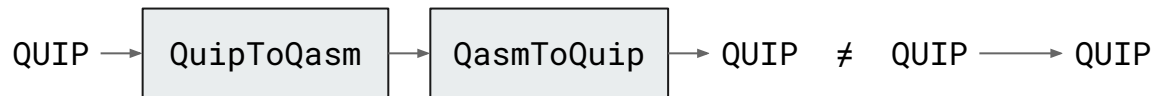
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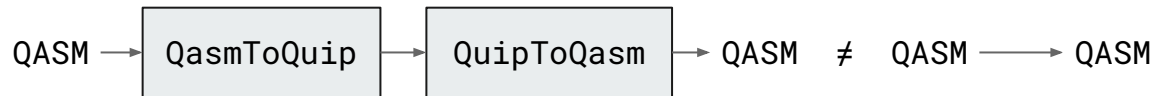
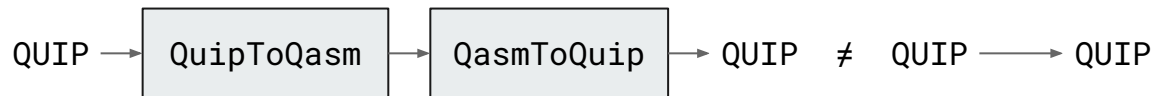
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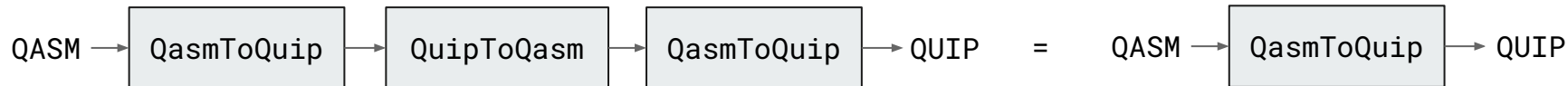
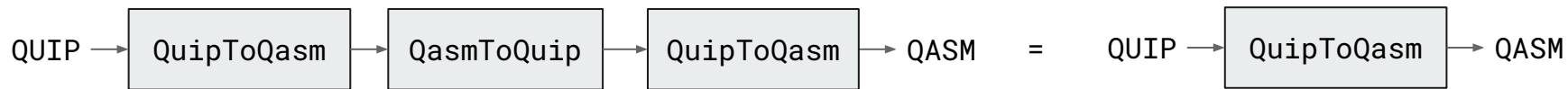
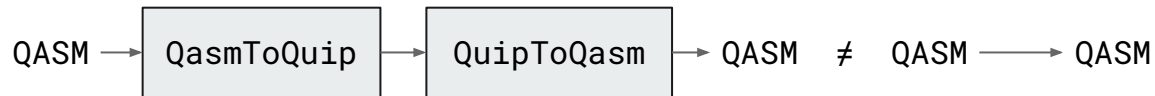
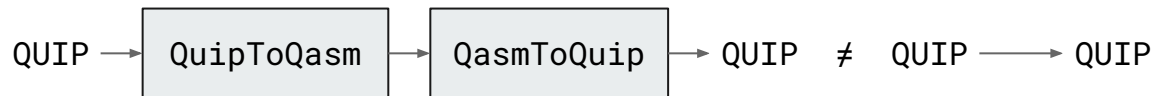
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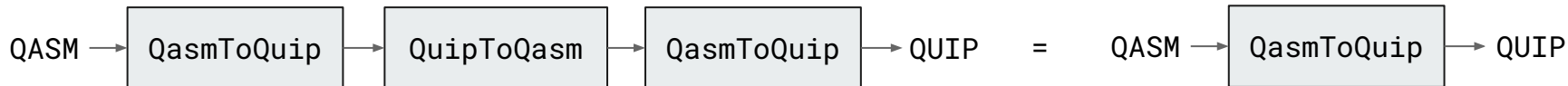
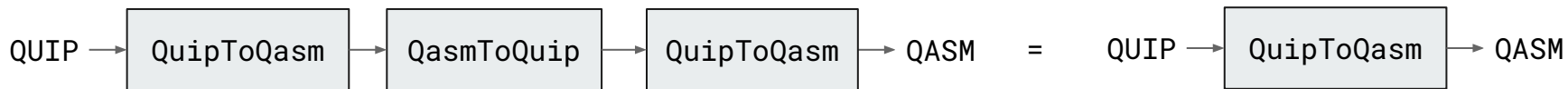
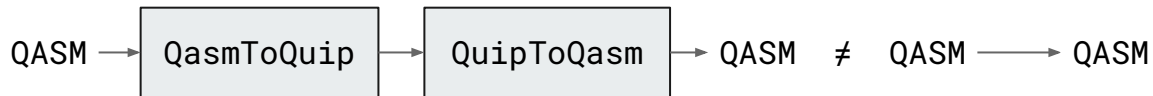
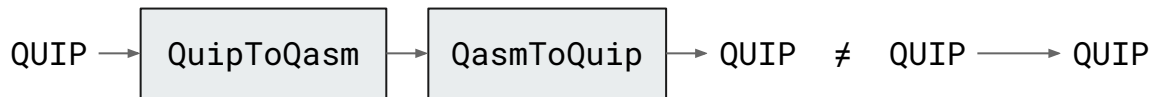
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Step 1: Round Translations (Idempotents)

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This set of requirements entail the **idempotence** of **round translations**.

We argue that these are reasonable properties for a transpiler to satisfy. They capture requirements such as round translates are **stable**, and **will not inflate** file size.

Step 2: A Closer Look at Translations

Q: Is there more structure to a transpilation stage?

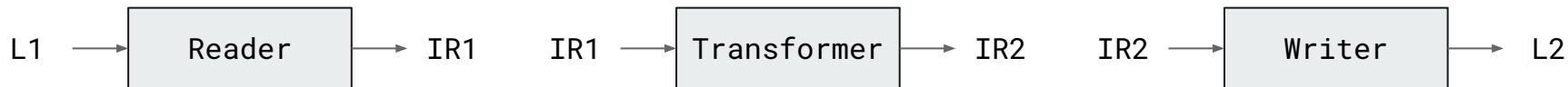


Step 2: A Closer Look at Translations

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We know from compiler design that **operating directly on source text is a bad design pattern...** at the very least, we should have a “**reader**” (lexer + parser) and a “**writer**” (code generation phase).

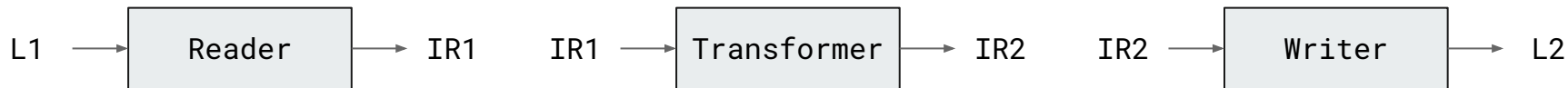


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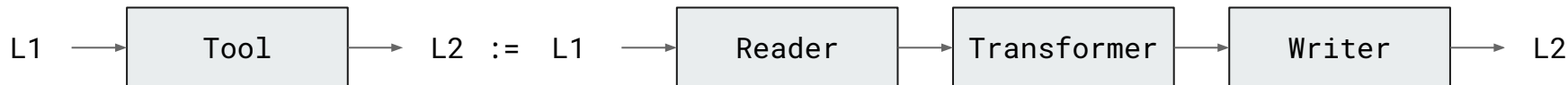
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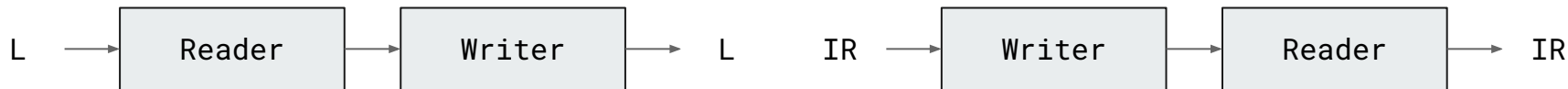


This yields the following definition:



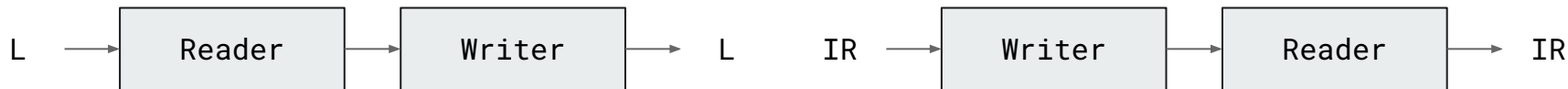
Step 3: Read-What-You-Write (Retracts)

Q: How should readers and writers interact?



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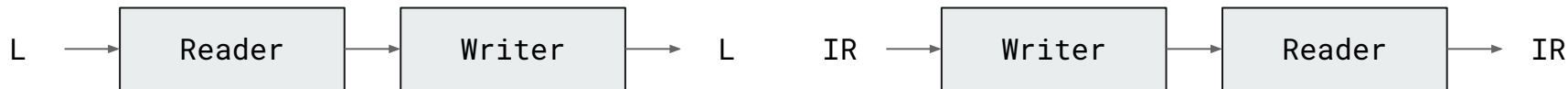
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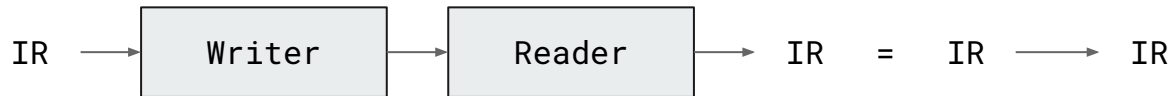
It is possible that **many syntactic structures** map to the same internal representation (IR). For example, pairs of inverse modifiers cancel out in OpenQASM 3. However, each IR statement can have a **canonical representation**.

Step 3: Read-What-You-Write (Retracts)

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It is possible that **many syntactic structures** map to the same internal representation (IR). For example, pairs of inverse modifiers cancel out in OpenQASM 3. However, each IR statement can have a **canonical representation**.



Categorically, this says that the reader should provide a **retraction** (left-inverse) for the writer.

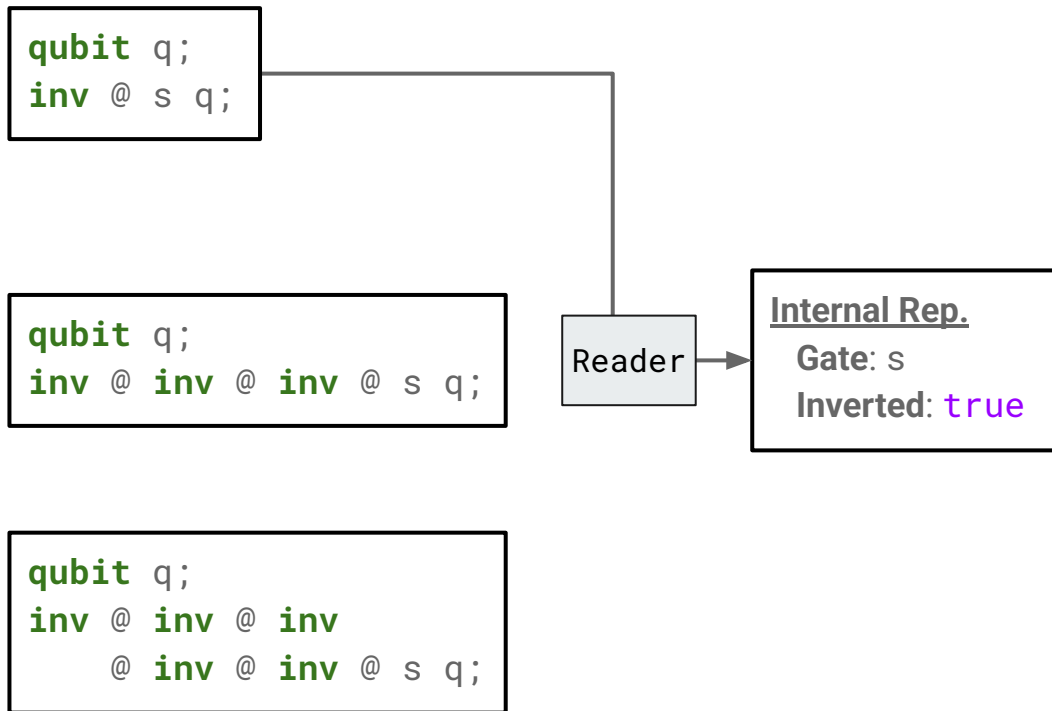
An Example Retraction With OpenQASM

```
qubit q;  
inv @ s q;
```

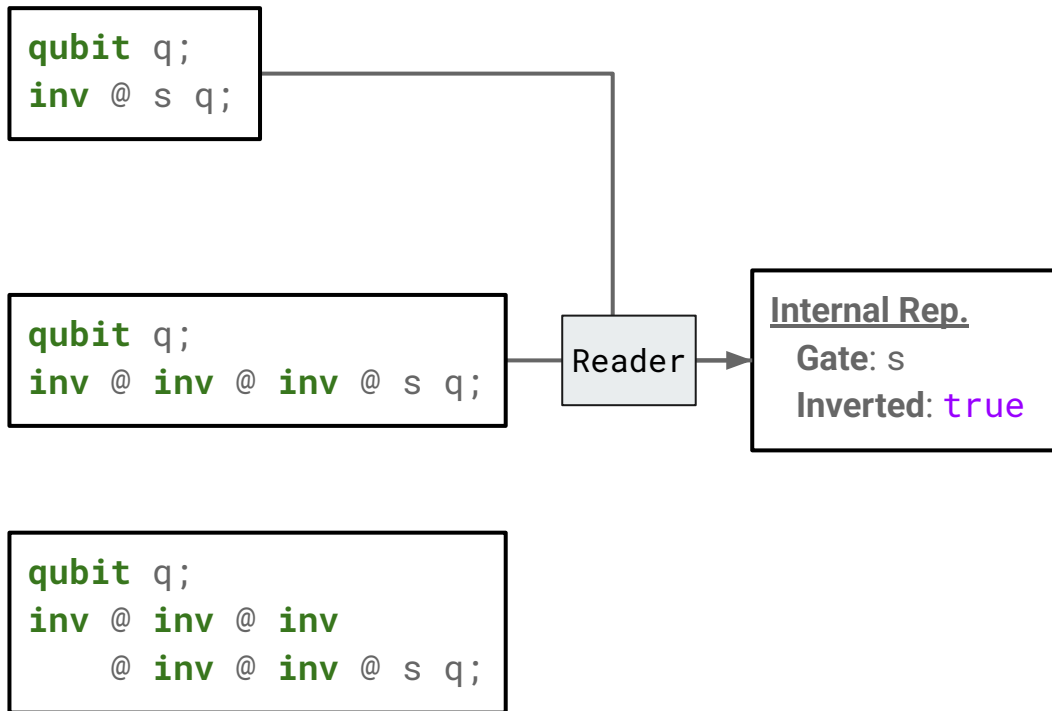
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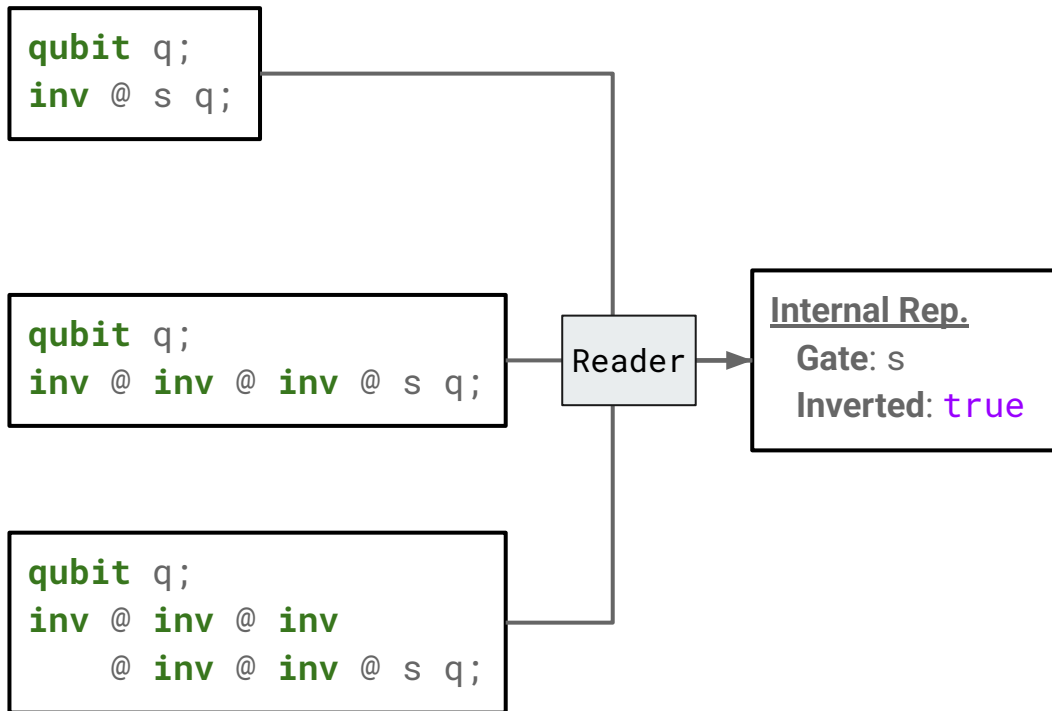
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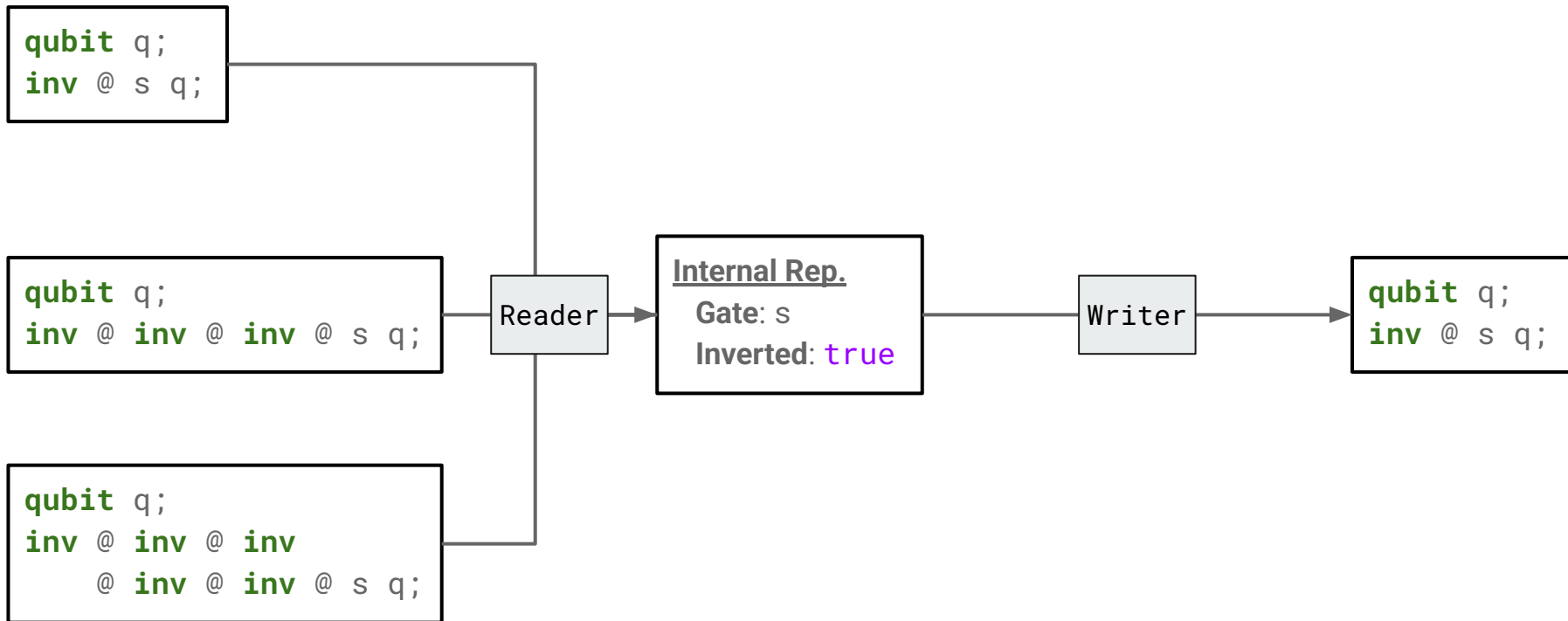
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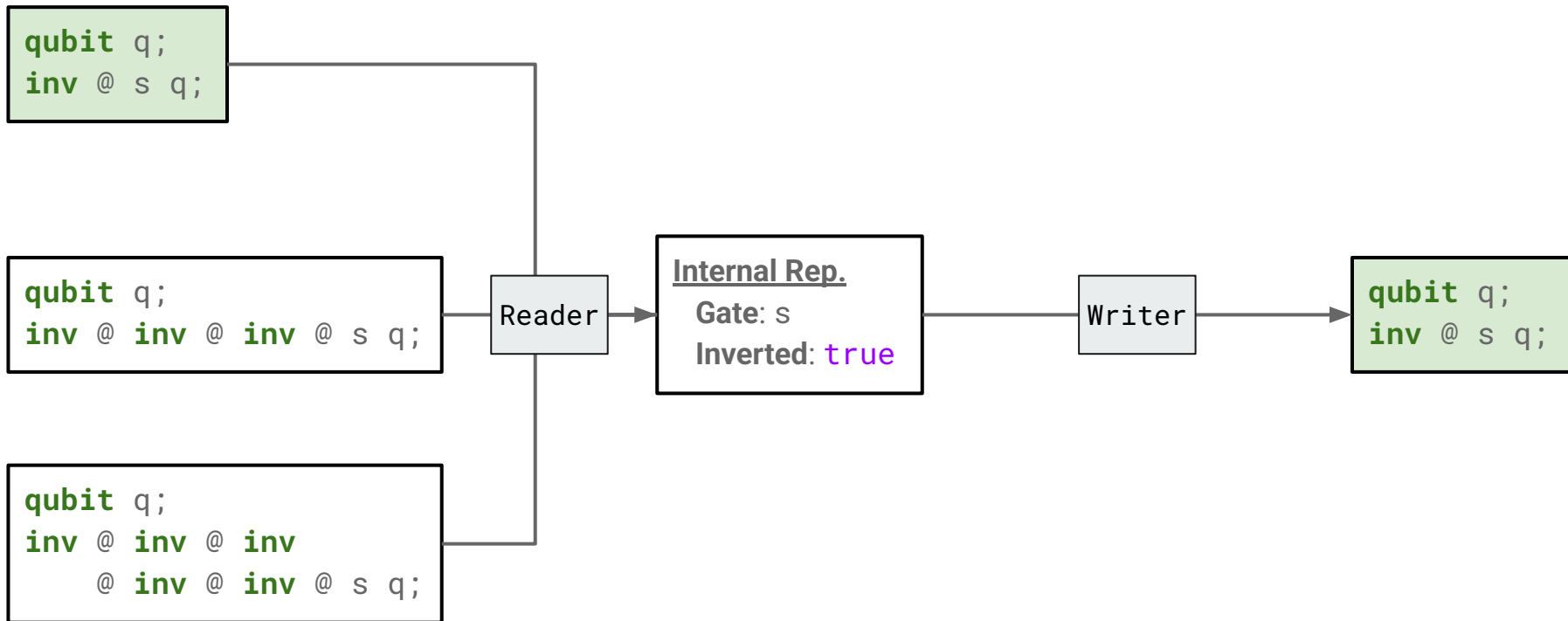
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Some Decompositions

Rewriting for a Quantum Transpiler?



OpenQASM and Quipper Lack Feature Parity

Notation	Matrix	QUIP	QASM3	QASM2
	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	✓	✓	✓
	$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$	✓	✓	✓
	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	✓	✓	✓
	$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$	✓	✓	✓
	$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$	✓	✓	✓
	$\begin{bmatrix} 1 & 0 \\ 0 & -i \end{bmatrix}$	✓	✓	✓
	$\begin{bmatrix} 1 & 0 \\ 0 & \omega \end{bmatrix}$	✓	✓	✓
	$\begin{bmatrix} 1 & 0 \\ 0 & \omega^* \end{bmatrix}$	✓	✓	✓
	$\frac{1}{2} \begin{bmatrix} 1+i & 1-i \\ 1-i & 1+i \end{bmatrix}$	✓	✓	×
	$\begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}$	✓	×	×
	$\begin{bmatrix} \omega & 0 \\ 0 & \omega \end{bmatrix}$	✓	×	×
	$\frac{1}{2} \begin{bmatrix} -1+i & 1+i \\ -1+i & -1-i \end{bmatrix}$	✓	×	×

Notation	Matrix	QUIP	QASM3	QASM2
	$I \oplus X$	✓	✓	✓
	$I \oplus Y$	✓	✓	✓
	$I \oplus Z$	✓	✓	✓
	$I \oplus H$	✓	✓	✓
	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	✓	✓	×
	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	✓	×	×
	$I \oplus I \oplus X$	✓	✓	✓
	$I \oplus \text{SWAP}$	✓	✓	×

Notation	Matrix	QUIP	QASM3	QASM2
	$e^{-iZ\theta}$	✓	×	×
	$R_x(\theta)$	×	✓	✓
	$R_y(\theta)$	×	✓	✓
	$R_z(\theta)$	×	✓	✓
	$P(\theta)$	✓	✓	×
	See Appx. A.	×	✓	✓
	See Appx. A.	×	✓	✓
	See Appx. A.	×	✓	✓
	See Appx. A.	×	✓	✓

Notation	Matrix	QUIP	QASM3	QASM2
	$I \oplus R_x(\theta)$	×	✓	×
	$I \oplus R_y(\theta)$	×	✓	×
	$I \oplus R_z(\theta)$	×	✓	✓
	$I \oplus P(\theta)$	×	✓	×
	$e^{i\gamma} \cdot U(\theta, \rho, \lambda)$	×	✓	×

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	$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$	✓	✓	✓
	$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$	✓	✓	✓
	$\begin{bmatrix} 1 & 0 \\ 0 & -i \end{bmatrix}$	✓	✓	✓
	$\begin{bmatrix} 1 & 0 \\ 0 & \omega \end{bmatrix}$	✓	✓	✓
	$\begin{bmatrix} 1 & 0 \\ 0 & \omega^* \end{bmatrix}$	✓	✓	✓
	$\frac{1}{2} \begin{bmatrix} 1+i & 1-i \\ 1-i & 1+i \end{bmatrix}$	✓	✓	×
	$\begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}$	✓	×	×
	$\begin{bmatrix} \omega & 0 \\ 0 & \omega \end{bmatrix}$	✓	×	×
	$\frac{1}{2} \begin{bmatrix} -1+i & 1+i \\ 1+i & -1-i \end{bmatrix}$	✓	×	×

Notation	Matrix	QUIP	QASM3	QASM2
	$I \oplus X$	✓	✓	✓
	$I \oplus Y$	✓	✓	✓
	$I \oplus Z$	✓	✓	✓
	$I \oplus H$	✓	✓	✓
	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	✓	✓	×
	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	✓	×	×
	$I \oplus I \oplus X$	✓	✓	✓
	$I \oplus \text{SWAP}$	✓	✓	×

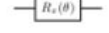
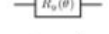
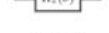
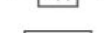
Notation	Matrix	QUIP	QASM3	QASM2
	$e^{-iZ\theta}$	✓	×	×
	$R_x(\theta)$	×	✓	✓
	$R_y(\theta)$	×	✓	✓
	$R_z(\theta)$	×	✓	✓
	$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{bmatrix}$	✓	✓	×
	See Appx. A.	×	✓	✓
	See Appx. A.	×	✓	✓
	See Appx. A.	×	✓	✓
	See Appx. A.	×	✓	✓

Notation	Matrix	QUIP	QASM3	QASM2
	$I \oplus R_x(\theta)$	×	✓	×
	$I \oplus R_y(\theta)$	×	✓	×
	$I \oplus R_z(\theta)$	×	✓	✓
	$I \oplus P(\theta)$	×	✓	×
	$e^{i\gamma} \cdot U(\theta, \rho, \lambda)$	×	✓	×

OpenQASM and Quipper Lack Feature Parity

Notation	Matrix	QUIP	QASM3	QASM2
	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	✓	✓	✓
	$\begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$	✓	✓	✓
	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	✓	✓	✓
	$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$	✓	✓	✓
	$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$	✓	✓	✓
	$\begin{bmatrix} 1 & 0 \\ 0 & -i \end{bmatrix}$	✓	✓	✓
	$\begin{bmatrix} 1 & 0 \\ 0 & \omega \end{bmatrix}$	✓	✓	✓
	$\begin{bmatrix} 1 & 0 \\ 0 & \omega^* \end{bmatrix}$	✓	✓	✓
	$\frac{1}{2} \begin{bmatrix} 1+i & 1-i \\ 1-i & 1+i \end{bmatrix}$	✓	✓	×
	$\begin{bmatrix} 0 & i \\ i & 0 \end{bmatrix}$	✓	×	×
	$\begin{bmatrix} \omega & 0 \\ 0 & \omega \end{bmatrix}$	✓	×	×
	$\frac{1}{2} \begin{bmatrix} -1+i & 1+i \\ -1+i & -1-i \end{bmatrix}$	✓	×	×

Notation	Matrix	QUIP	QASM3	QASM2
	$I \oplus X$	✓	✓	✓
	$I \oplus Y$	✓	✓	✓
	$I \oplus Z$	✓	✓	✓
	$I \oplus H$	✓	✓	✓
	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	✓	✓	×
	$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	✓	×	×
	$I \oplus I \oplus X$	✓	✓	✓
	$I \oplus \text{SWAP}$	✓	✓	×

Notation	Matrix	QUIP	QASM3	QASM2
	$e^{-iZ\theta}$	✓	×	×
	$R_x(\theta)$	×	✓	✓
	$R_y(\theta)$	×	✓	✓
	$R_z(\theta)$	×	✓	✓
	$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{bmatrix}$	✓	✓	×
	See Appx. A.	×	✓	✓
	See Appx. A.	×	✓	✓
	See Appx. A.	×	✓	✓
	See Appx. A.	×	✓	✓

Notation	Matrix	QUIP	QASM3	QASM2
	$I \oplus R_x(\theta)$	×	✓	×
	$I \oplus R_y(\theta)$	×	✓	×
	$I \oplus R_z(\theta)$	×	✓	✓
	$I \oplus P(\theta)$	×	✓	×
	$e^{i\gamma} \cdot U(\theta, \rho, \lambda)$	×	✓	×

Specific Decompositions for Specific Gates

Named gates without parameters typically require **one-off translations**. A part of this project was to bring together a list of these translations, in a “**Rewriting Compendium**”, so to speak.

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Some statistics about this compendium:

1. **Number of References:** 9
2. **Number of Rewrite Rules:** 52
3. **Number of Non-Elementary Relations:** 42

This compendium is nowhere near comprehensive, but we hope for it to be a **start for compiler design**.

Specific Decompositions for Specific Gates

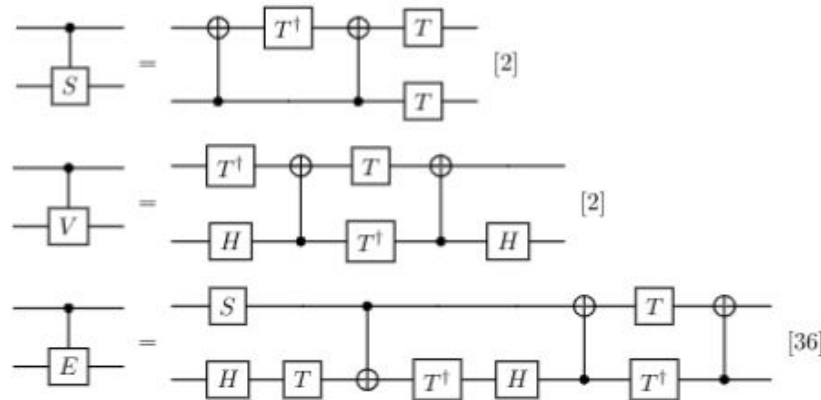
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Some Example Rewrite Rules:

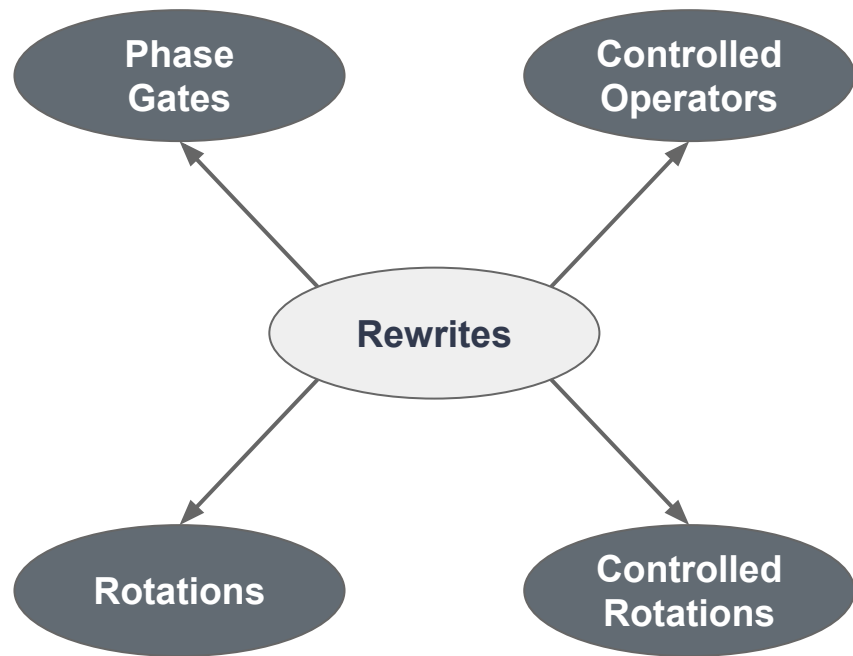


[2] M. Amy, D. Maslov, M. Mosca, M. Roetteler 2013.

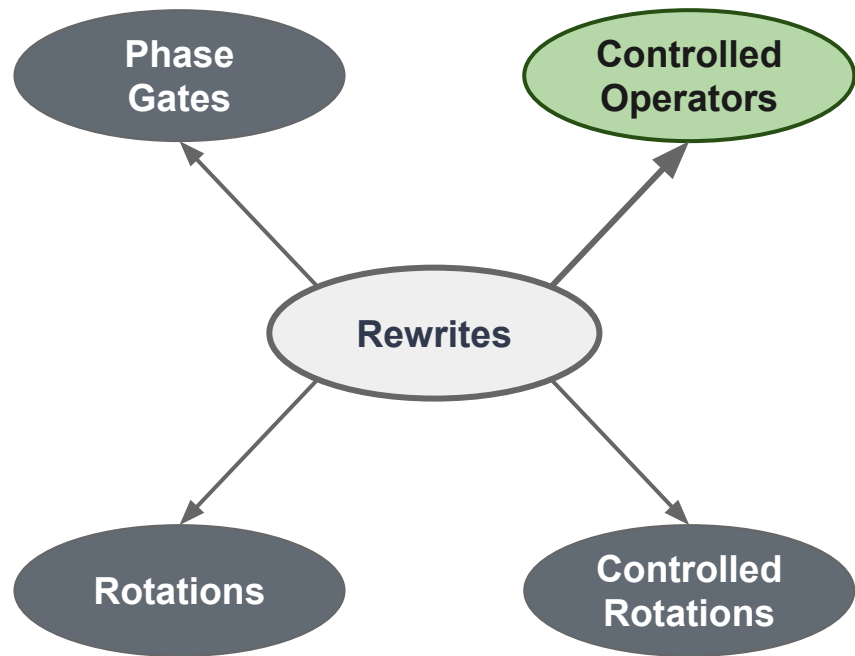
[36]

<https://www.mathstat.dal.ca/~selinger/quipper/doc/>

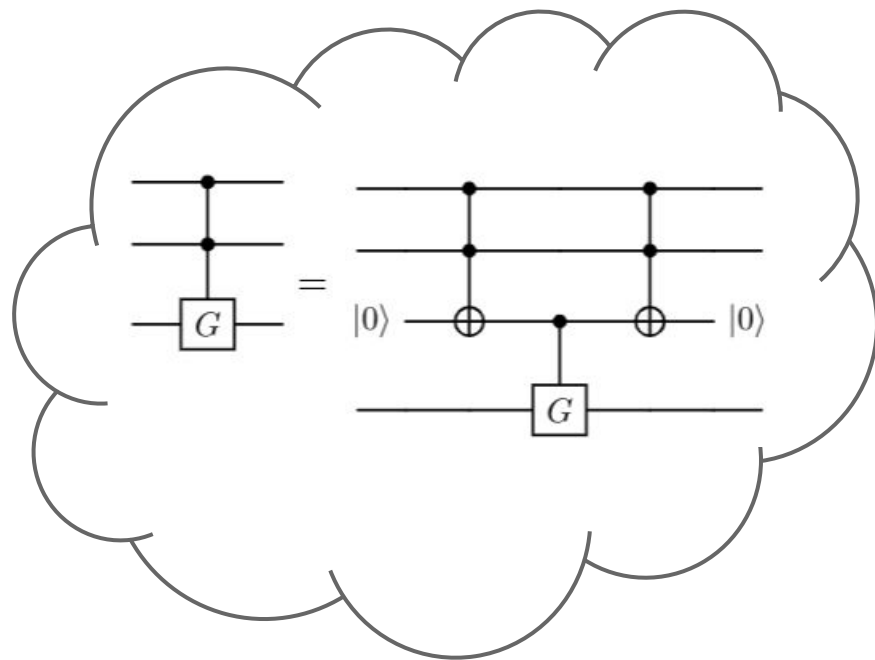
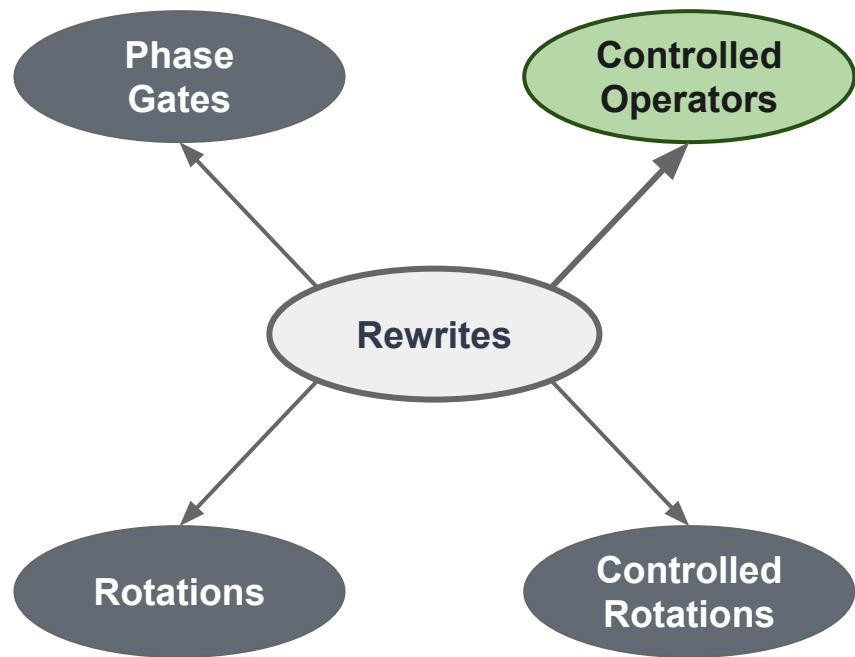
Wider Classes of Gate Decompositions



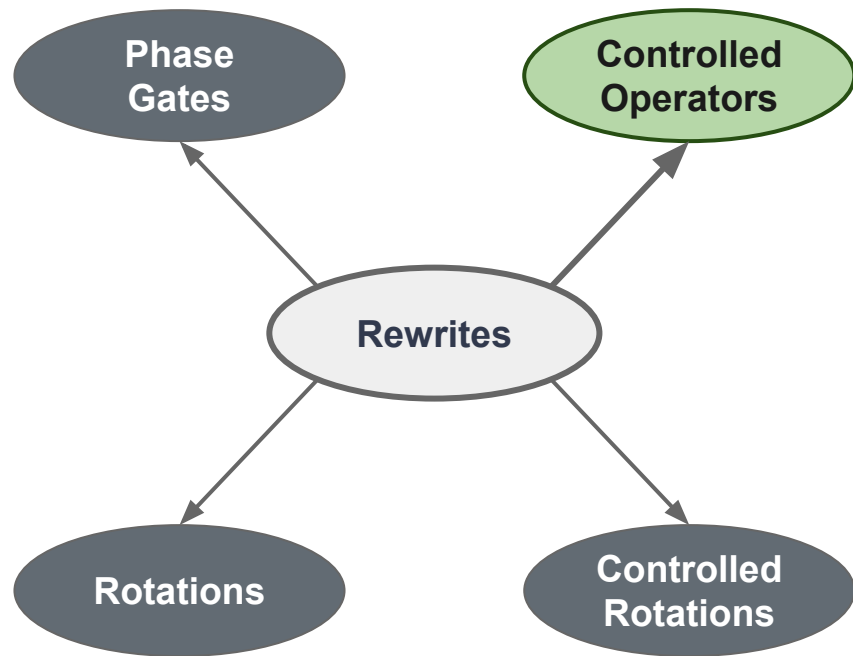
Wider Classes of Gate Decompositions



Wider Classes of Gate Decompositions



Wider Classes of Gate Decompositions



Given a unitary operator U such that:

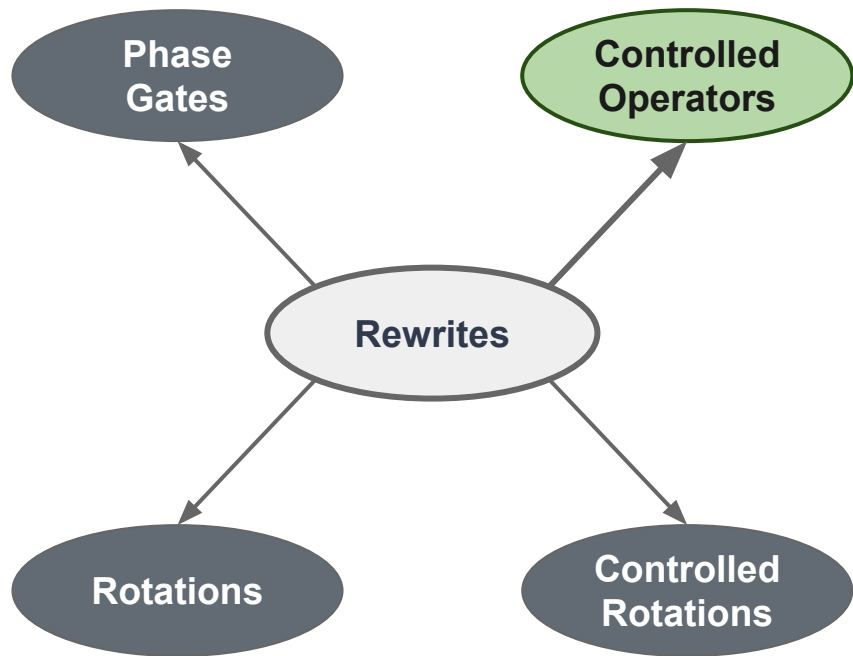
$$U |000\rangle = |\varphi_1\rangle |0\rangle$$

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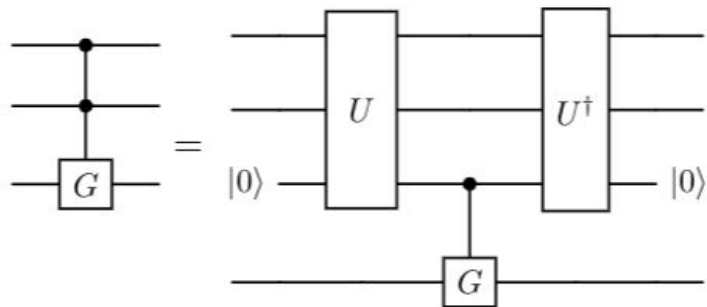
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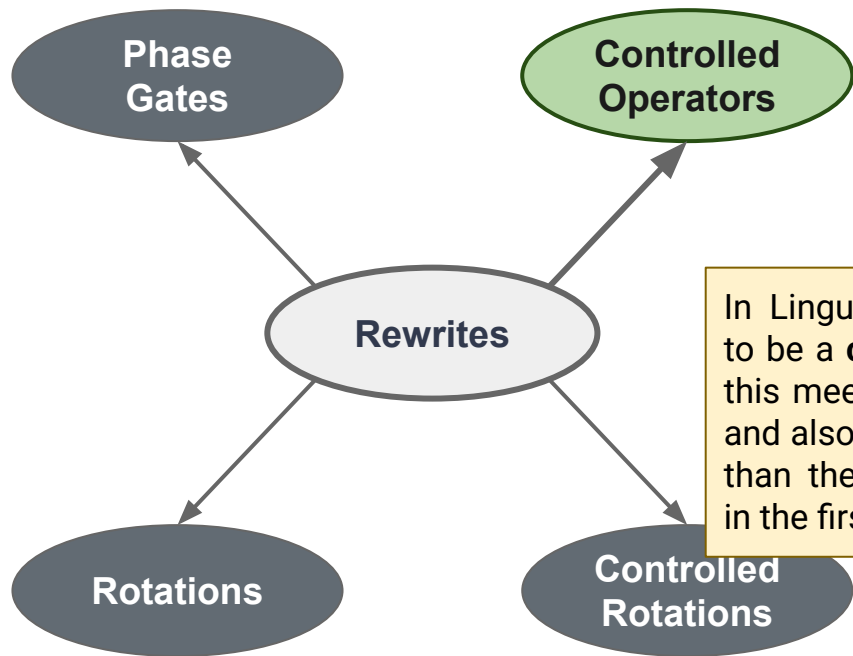
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[35] Selinger, 2013.

Wider Classes of Gate Decompositions

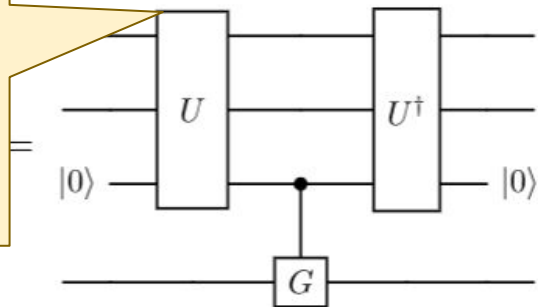


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[35] Selinger, 2013.

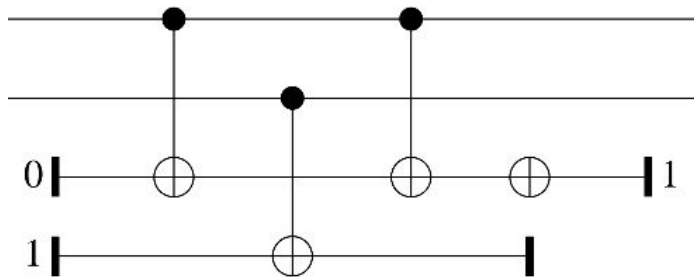
Ancilla Management

Translation from Quipper to OpenQASM



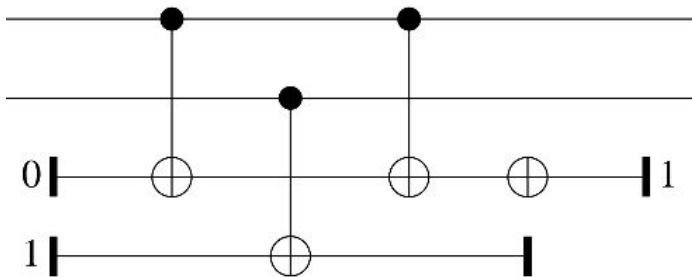
Simulating Ancillas in OpenQASM 3

Quipper circuits allow for **ancilla qubits**.



Simulating Ancillas in OpenQASM 3

Quipper circuits allow for **ancilla qubits**.



Ancillas are managed using the **following primitives**.

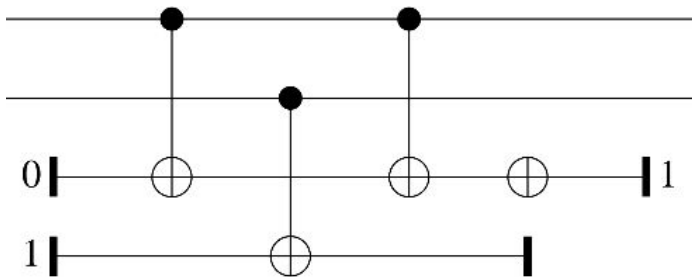
Term

Init

Discard

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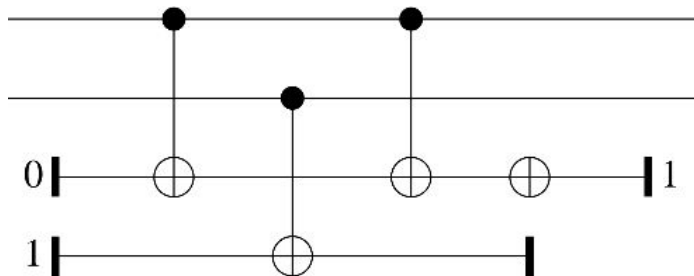
Init

Discard

Conversely, OpenQASM allocates qubits **up front**.

Simulating Ancillas in OpenQASM 3

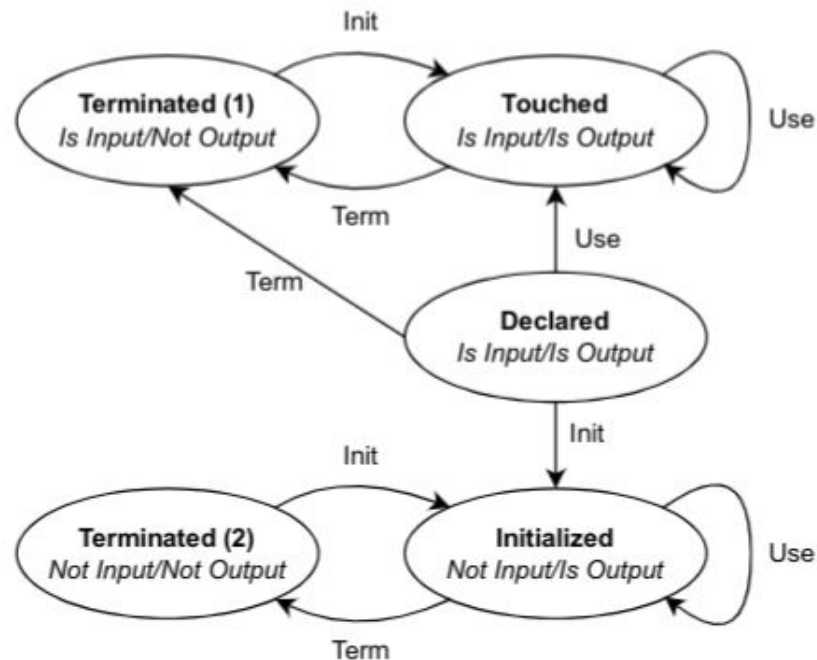
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Term Init Discard

Conversely, OpenQASM allocates qubits **up front**.



An Application and Other Features

Coming Back to the LSC Compiler



Back to the Motivating Example

Realistic Cost to Execute Practical Quantum Circuits using Direct Clifford+T Lattice Surgery Compilation

TYLER LEBLOND, Computational Sciences and Engineering Division

Oak Ridge National Laboratory, USA

CHRISTOPHER DEAN, Department of Mathematics and Statistics

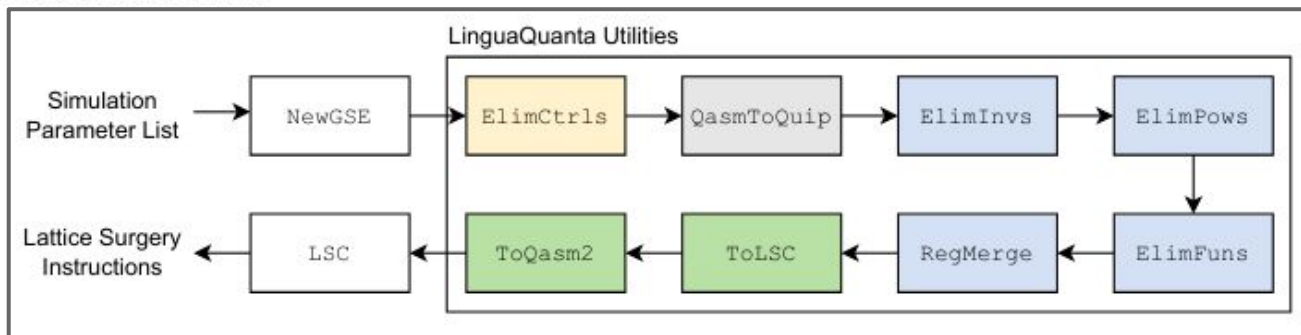
Dalhousie University, Canada

GEORGE WATKINS, Department of Computer Science

Aalto University, Finland

RYAN S. BENNINK, Computational Sciences and Engineering Division

Oak Ridge National Laboratory, USA



A List of Key Limitations

Ongoing Development:

1. Support for type conversion in Quipper.
2. Support for OpenQASM's user-defined gates.
3. Support for opaque gates.
4. Basic support for classical control.
5. Constant propagation for OpenQASM 3.
6. Bounded loop unwinding for OpenQASM 3.

Out-of-scope Features:

1. No-control flags in Quipper.
2. Dynamic lifting in Quipper.
3. Generalized controls in Quipper
4. OpenPulse and physical qubits in OpenQASM 3.
5. Floats and dynamic arrays in OpenQASM 3.

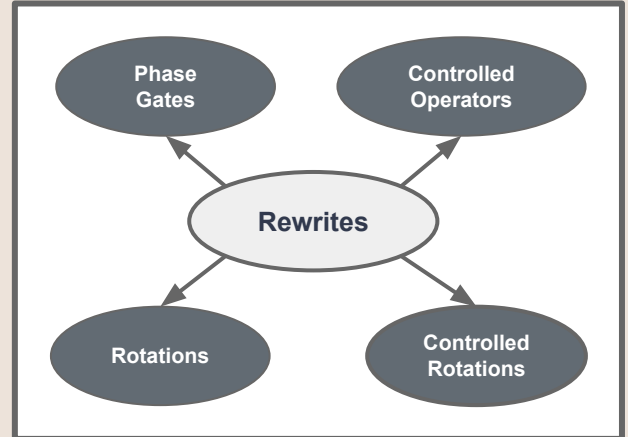
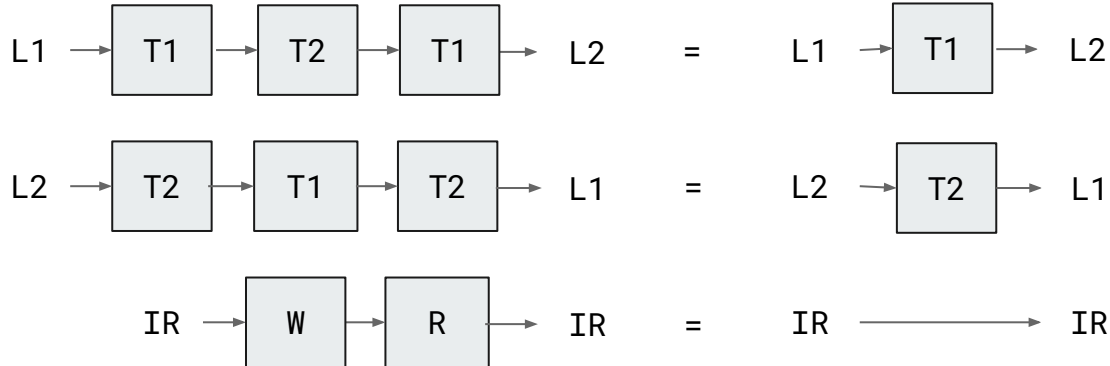
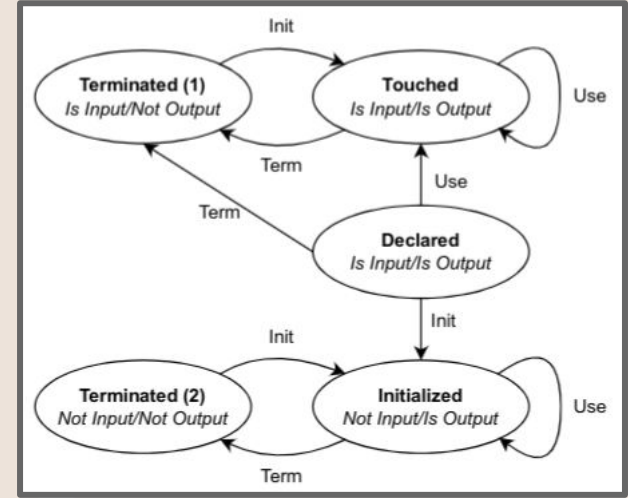
Future Research Directions

1. Categorical semantics for preprocessing in a transpilation pipeline.
2. Formal verification of LinguaQuanta and a general methodology.
3. Whitebox fuzzing of pipeline software through pipeline rewriting.
4. Empirical evaluations of LinguaQuanta (performance and conformance).

Questions?

Want to learn more? Contact: scott.wesley@dal.ca

Want to try LinguaQuanta? Visit: github.com/onestruggler/qasm-guipper



Appendix A

Sub-Languages, Lattices, and Pipelines

Future Work: Hierarchical Types

There are many **preprocessing** tools in LinguaQuanta.

ElimFuns

ElimPows

ElimInvs

Future Work: Hierarchical Types

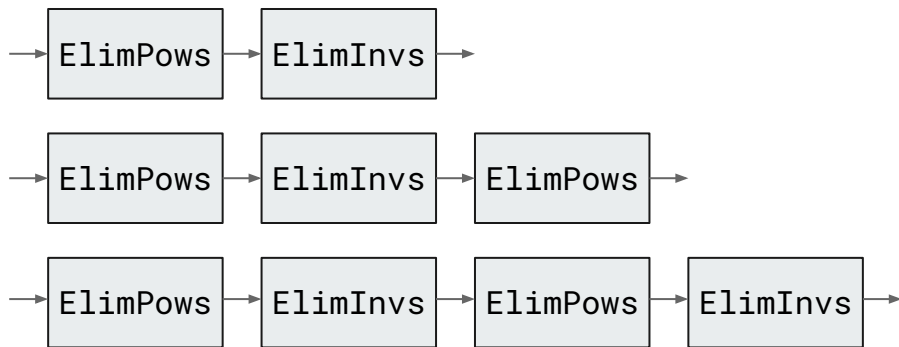
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ElimFuns

ElimPows

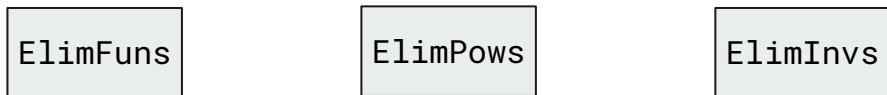
ElimInvs

One could expect these pipelines to be **equivalent**.



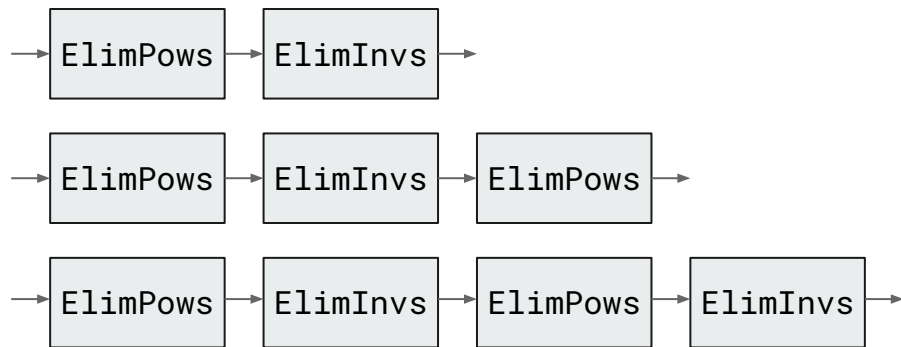
Future Work: Hierarchical Types

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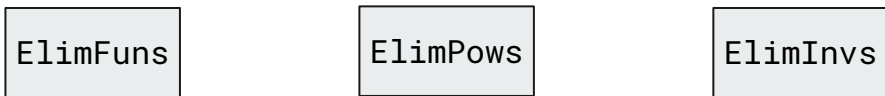
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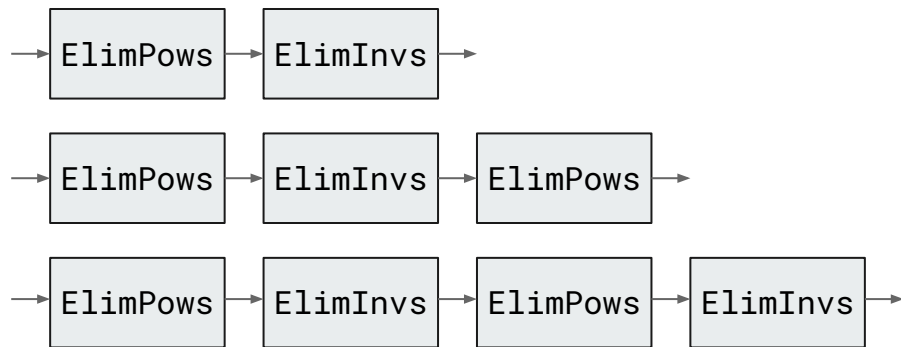


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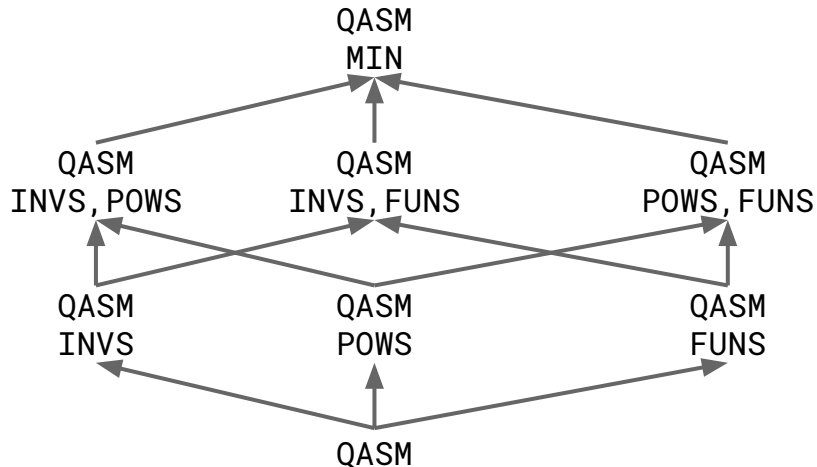


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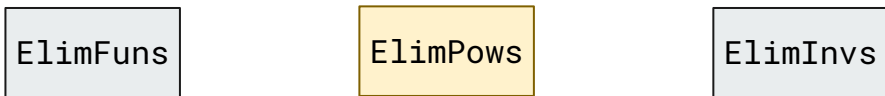
Though **commutativity** seems **too strong** a property.

Our proposed solution is some sort of **subtyping**.

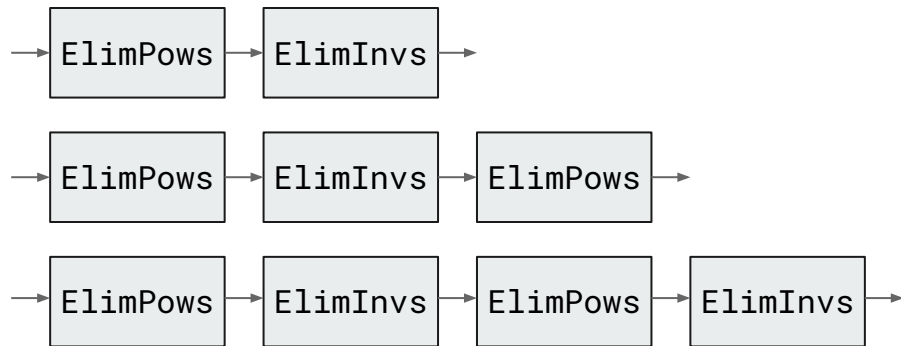


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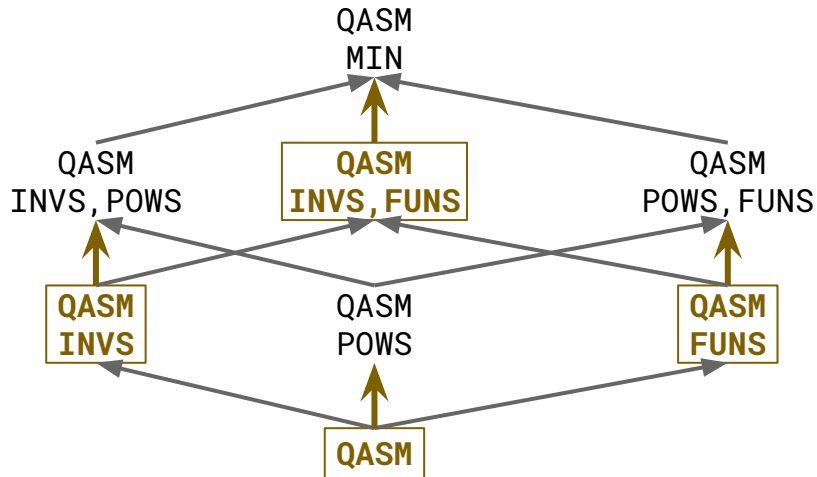


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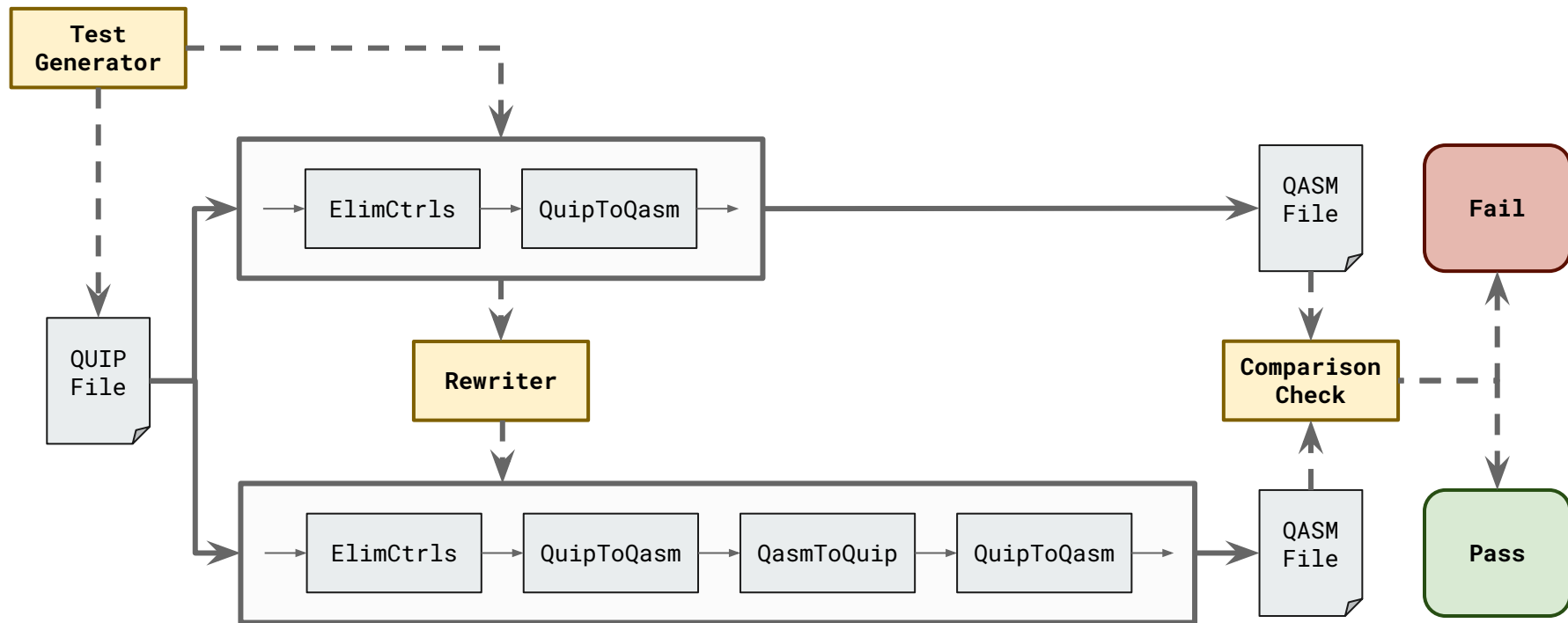
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Appendix B

Whitebox Fuzzing and Pipeline Software

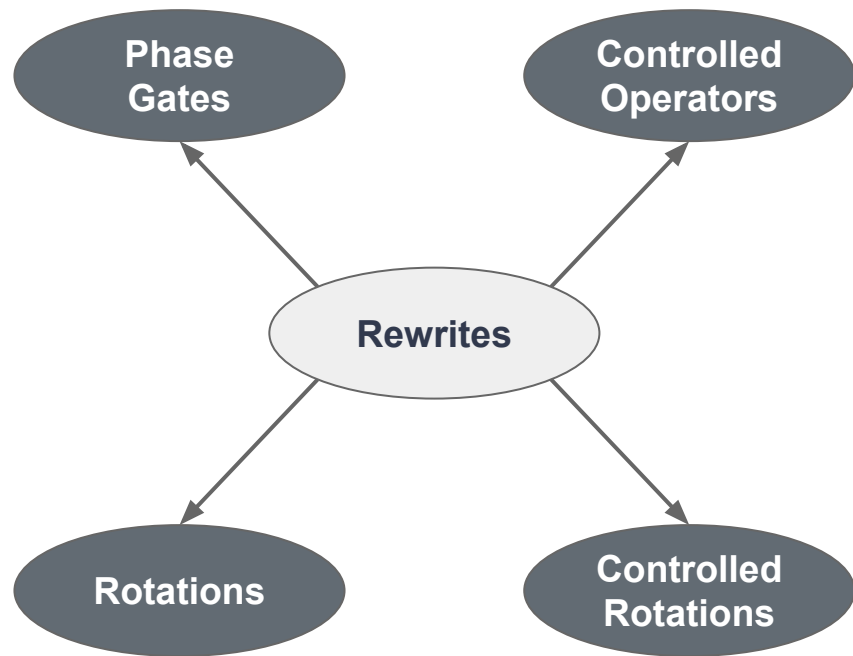
Future Work: Whitebox Fuzzing



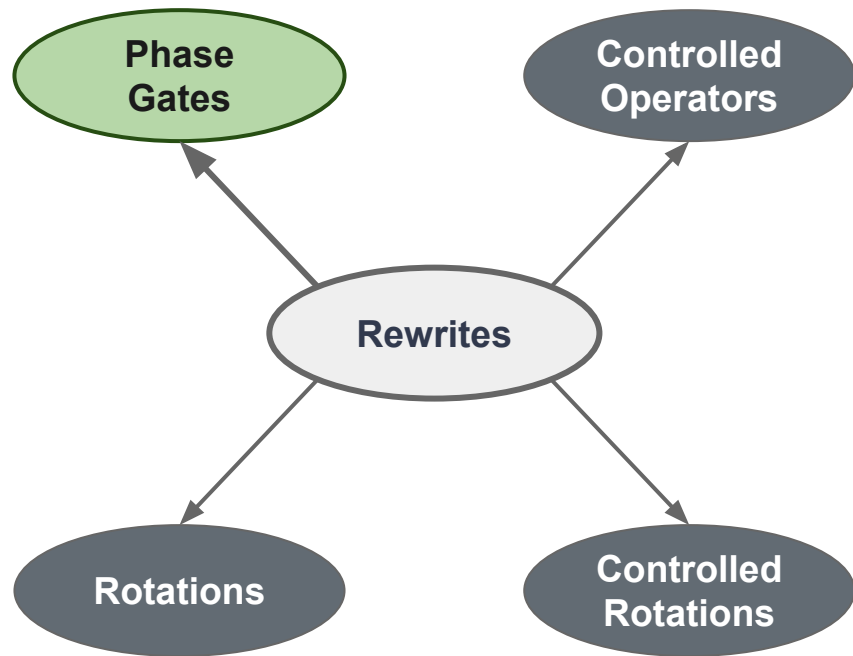
Appendix C

More Gate Decomposition Templates

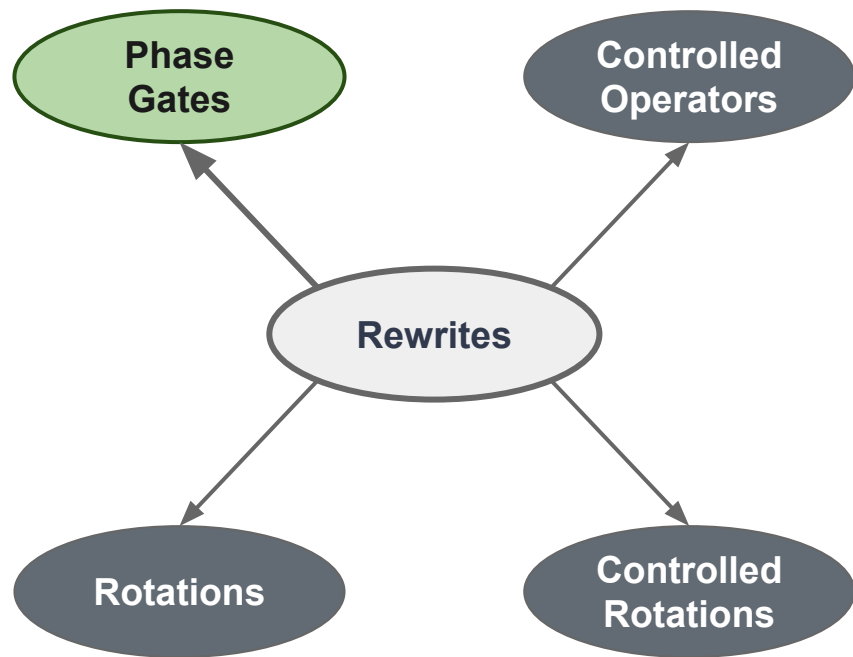
Wider Classes of Gate Decompositions



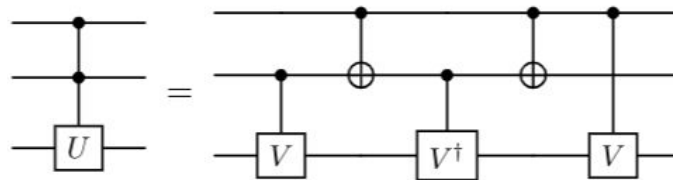
Wider Classes of Gate Decompositions



Wider Classes of Gate Decompositions

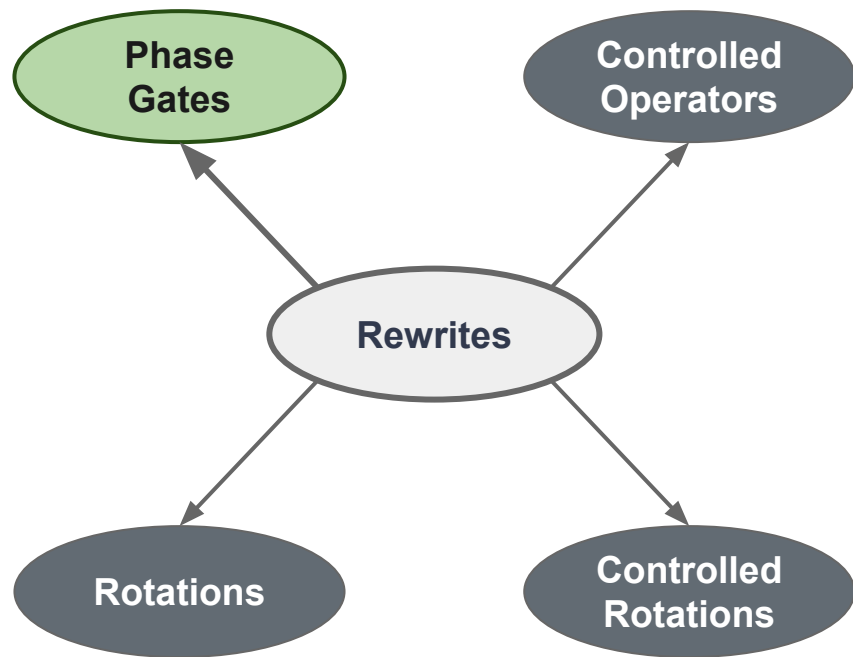


Assume that V is a unitary and $U = V^2$.

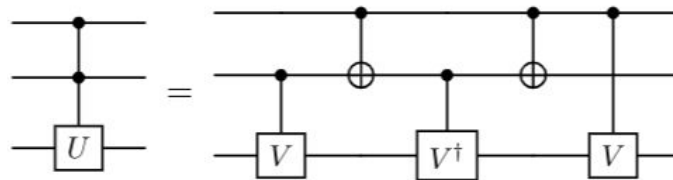


[4] A. Barenco et al., 1995.

Wider Classes of Gate Decompositions

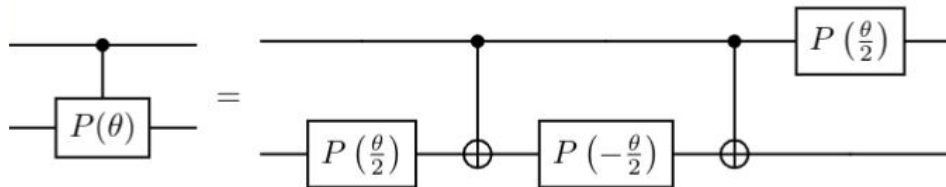


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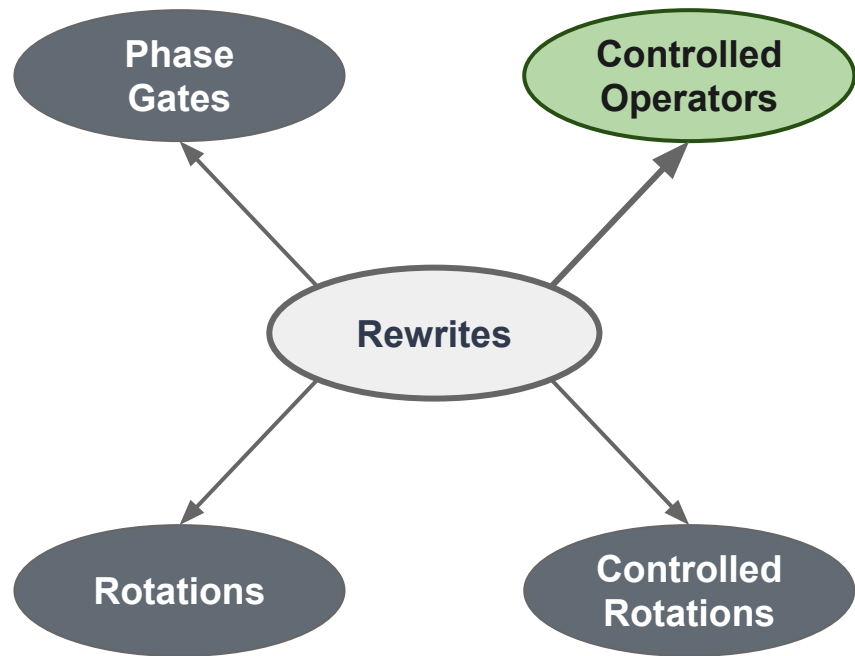


[4] A. Barenco et al., 1995.

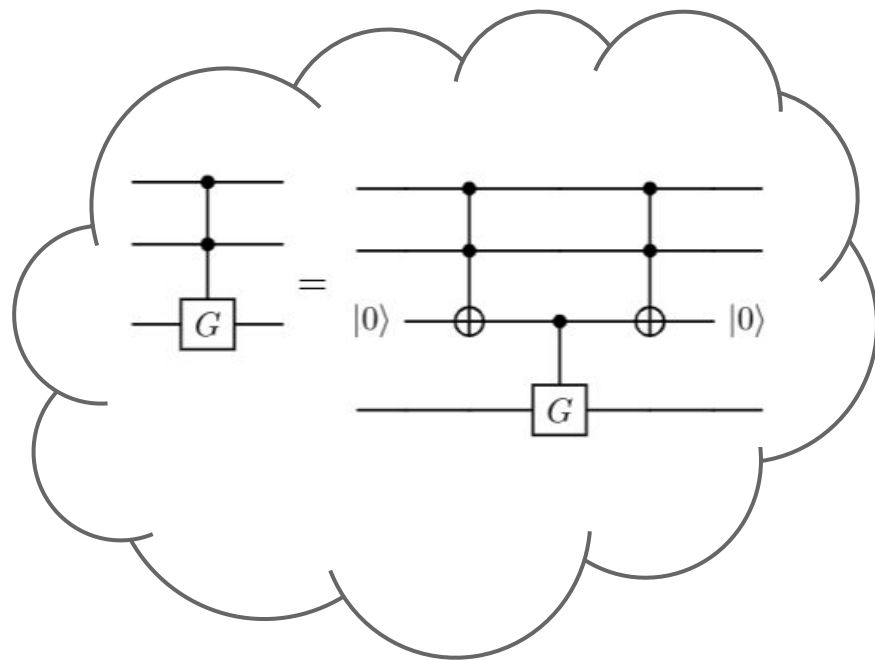
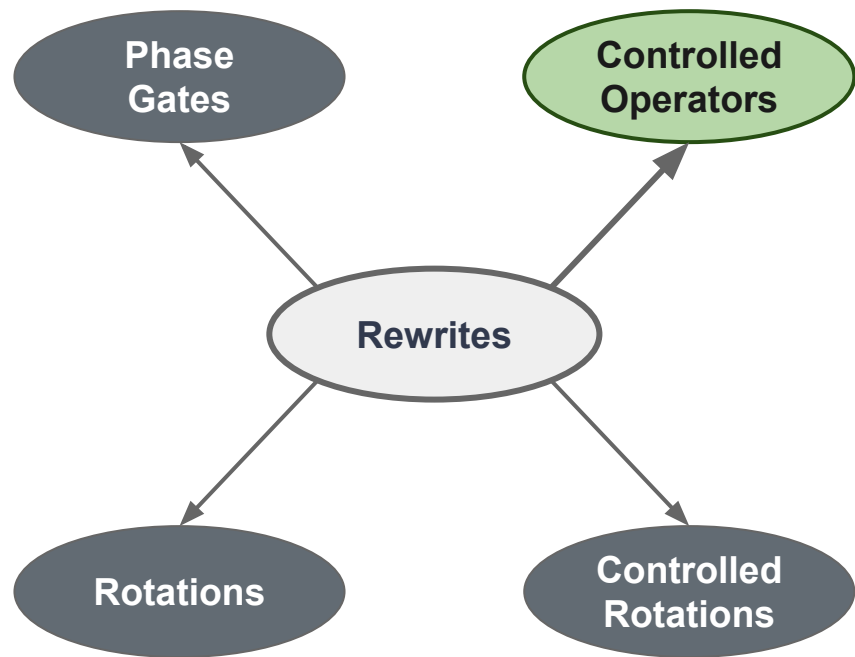
The $P(\theta)$ gate in OpenQASM is a controlled phase.



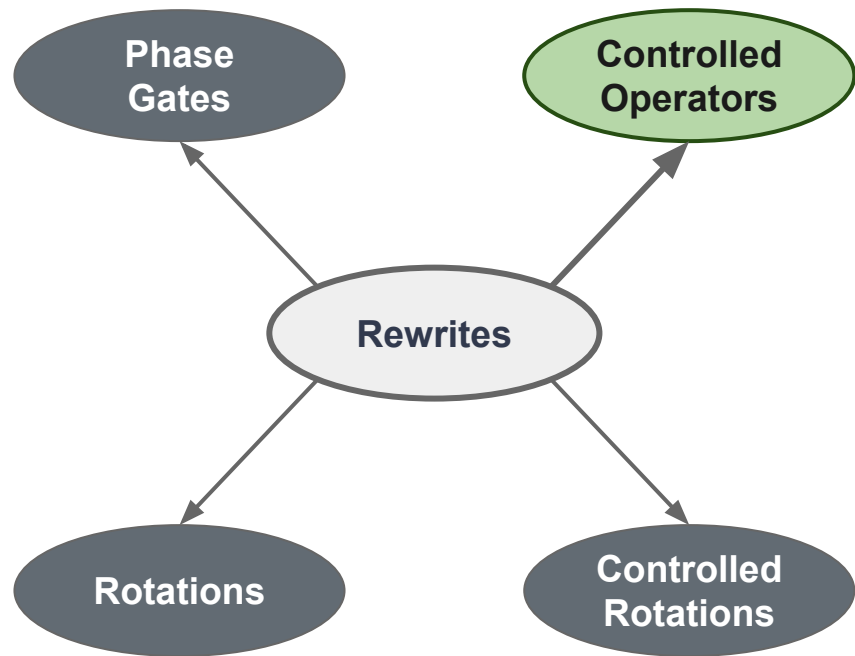
Wider Classes of Gate Decompositions



Wider Classes of Gate Decompositions



Wider Classes of Gate Decompositions



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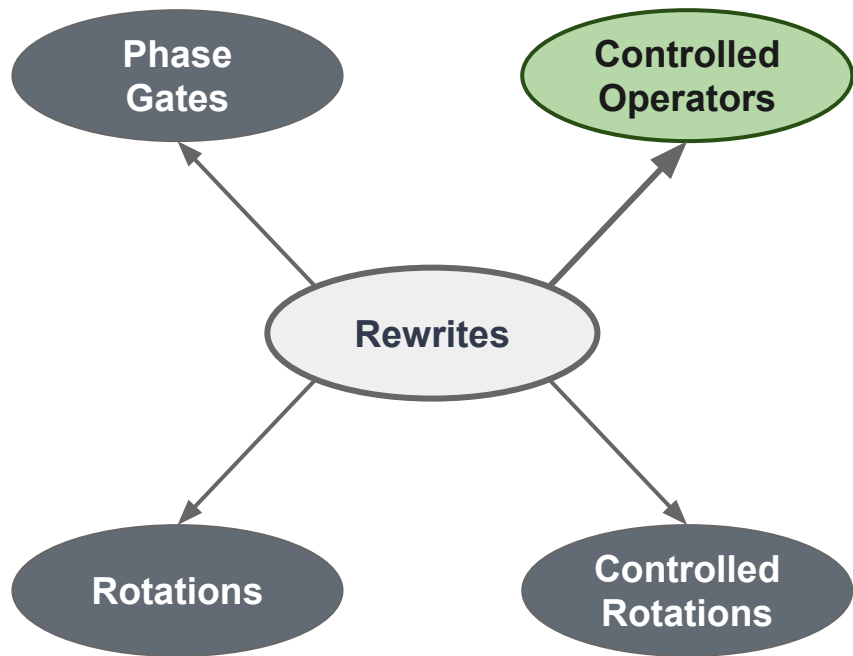
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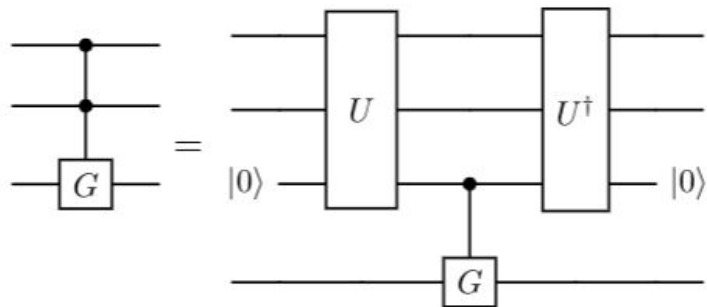
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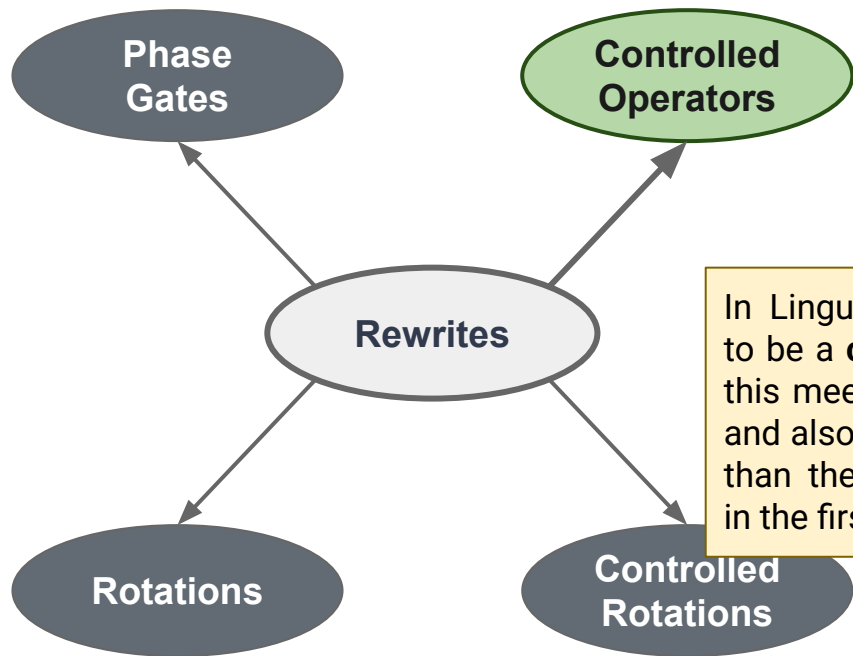
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The following equation necessarily hold:



[35] Selinger, 2013.

Wider Classes of Gate Decompositions

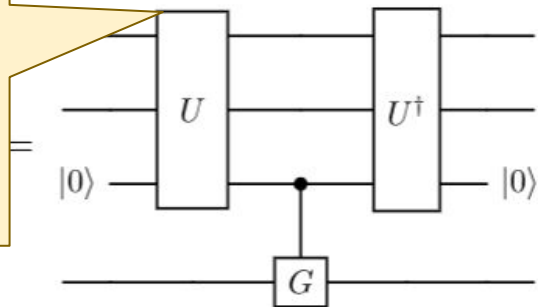


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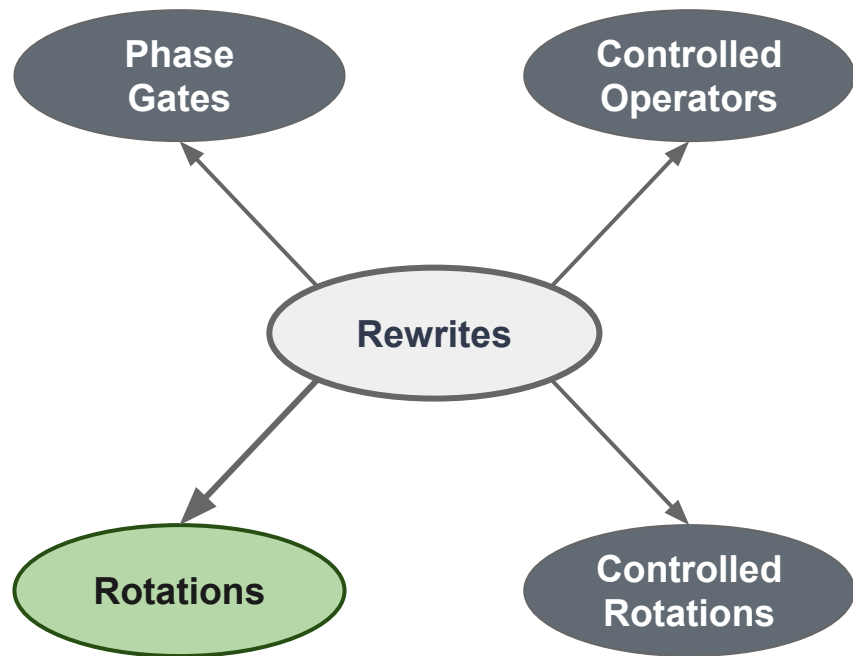
The following equation necessarily hold:

In LinguaQuanta, we take U to be a **controlled iX** gate as this meets the specifications and also has a **lower T-count** than the Toffoli gate shown in the first decomposition.

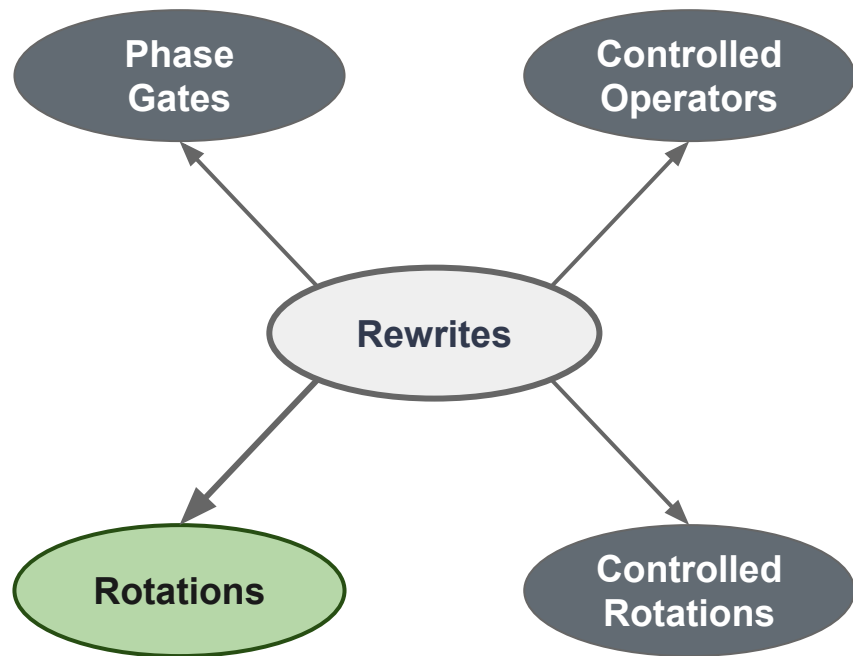


[35] Selinger, 2013.

Wider Classes of Gate Decompositions



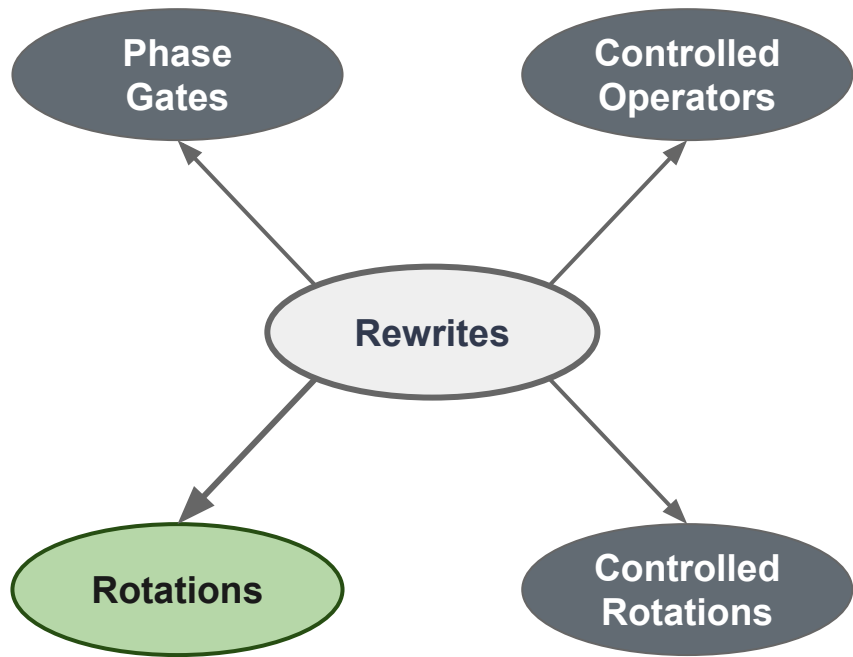
Wider Classes of Gate Decompositions



Recall the following fact about matrix exponentials.

$$e^{U^\dagger V U} = U^\dagger e^V U$$

Wider Classes of Gate Decompositions



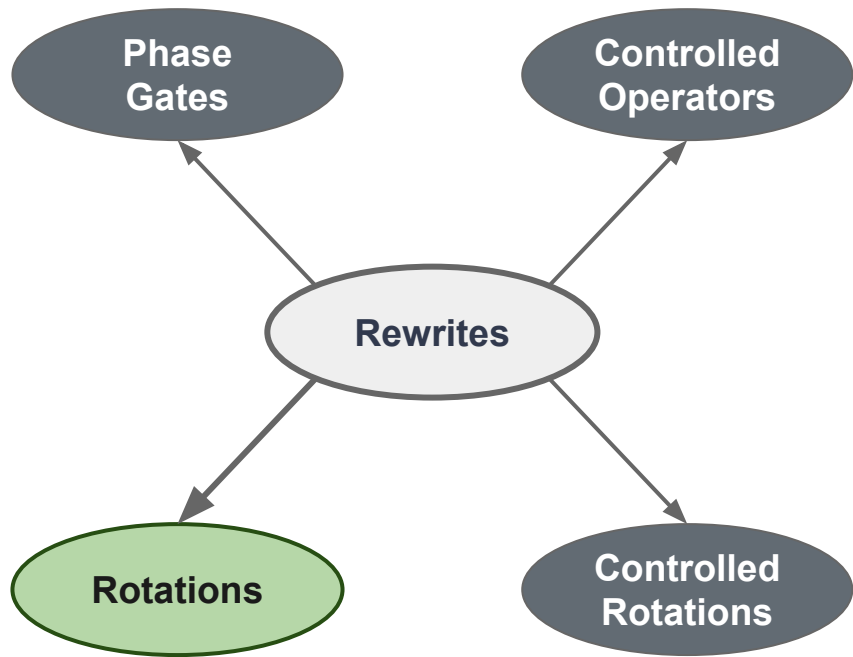
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Standard rotations are also matrix exponentials.

$$R_A(\theta) = e^{iA\theta} \quad R_B(\theta) = e^{iB\theta}$$

Wider Classes of Gate Decompositions



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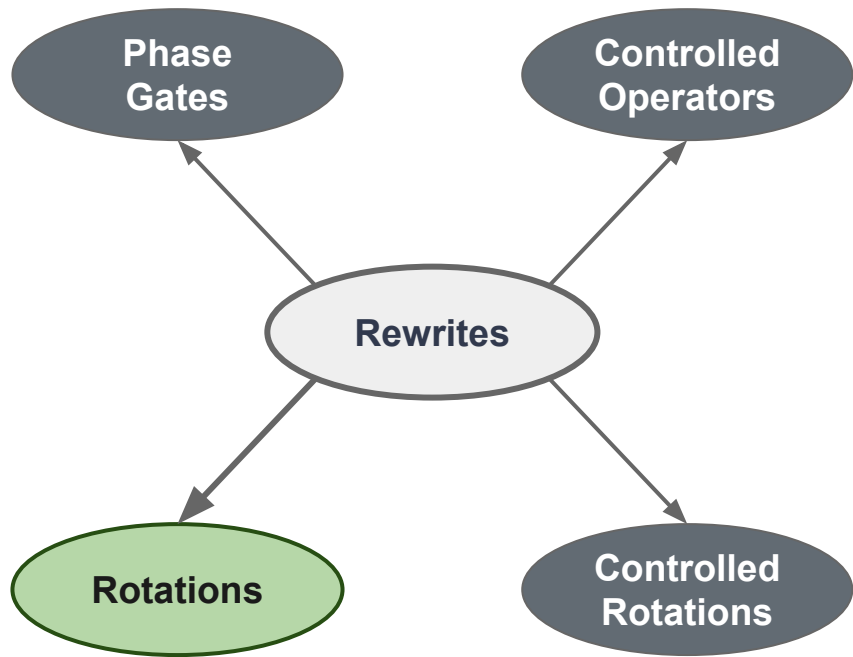
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Wider Classes of Gate Decompositions



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Then we obtain the following equality.

$$\text{---} \boxed{R_B(\theta)} \text{---} = \text{---} \boxed{V} \text{---} \boxed{R_A(\theta)} \text{---} \boxed{V^\dagger} \text{---}$$

Wider Classes of Gate Decompositions

Phase

Controlled

Recall the following fact about matrix exponentials.

$$\text{---} \boxed{Rx(\theta)} \text{---} = \text{---} \boxed{H} \text{---} \boxed{\exp Z\left(\frac{\theta}{2}\right)} \text{---} \boxed{H} \text{---}$$

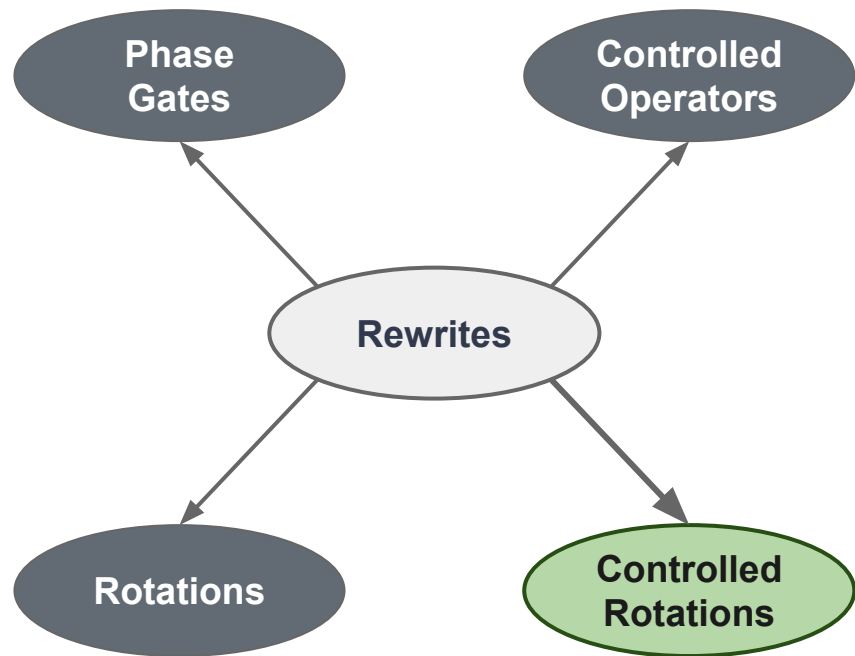
$$\text{---} \boxed{Ry(\theta)} \text{---} = \text{---} \oplus \text{---} \boxed{S} \text{---} \boxed{H} \text{---} \boxed{\exp Z\left(\frac{\theta}{2}\right)} \text{---} \boxed{H} \text{---} \boxed{S^\dagger} \text{---} \oplus \text{---}$$

Rotations

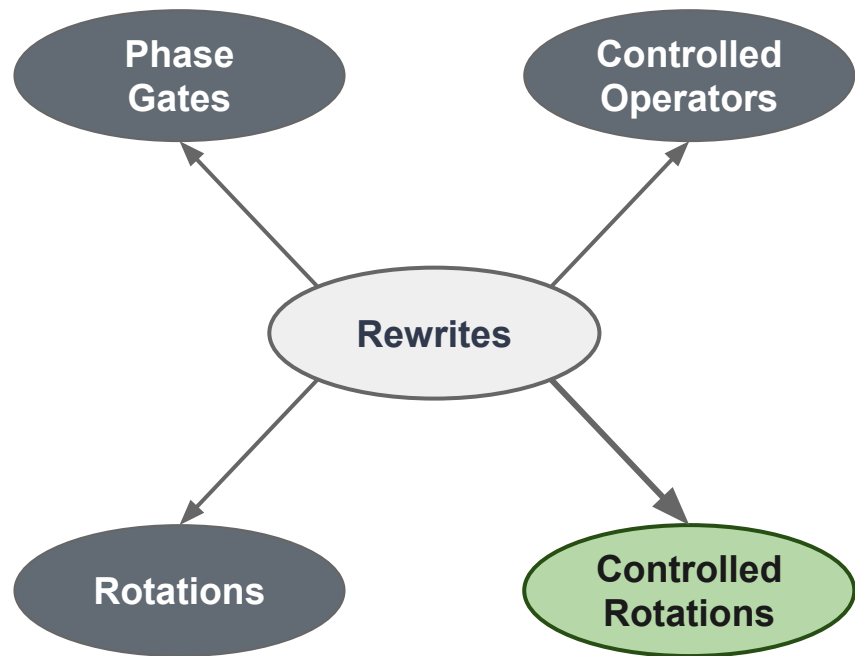
Rotations

$$\text{---} \boxed{R_B(\theta)} \text{---} = \text{---} \boxed{V} \text{---} \boxed{R_A(\theta)} \text{---} \boxed{V^\dagger} \text{---}$$

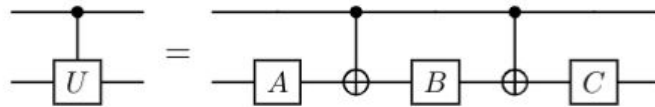
Wider Classes of Gate Decompositions



Wider Classes of Gate Decompositions

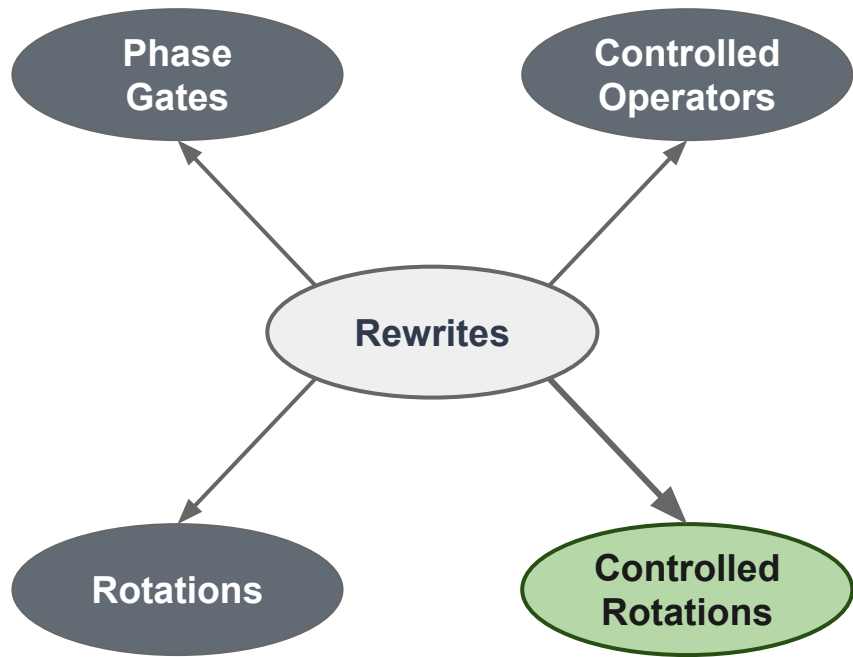


Assume that $U = CXBXA$ and $CBA = I$.

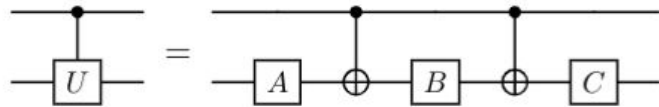


[4] A. Barenco et al., 1995.

Wider Classes of Gate Decompositions



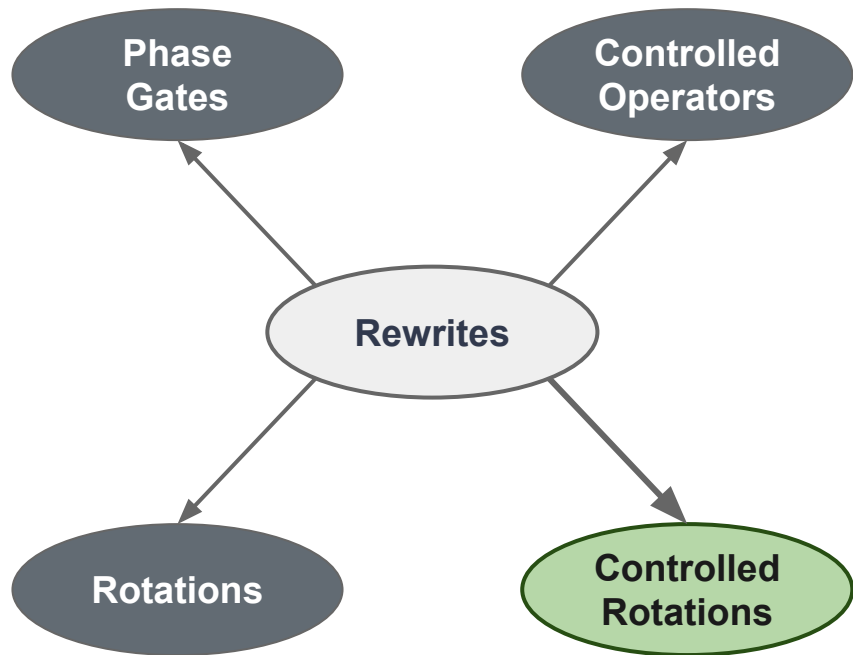
Assume that $U = CXBXA$ and $CBA = I$.



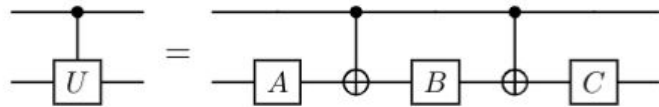
[4] A. Barenco et al., 1995.

Assume that D is self-inverse and R is a rotation satisfying $(DXD)R(\theta)(DXD) = R(-\theta)$ and $R(\theta)R(-\theta) = I$.

Wider Classes of Gate Decompositions



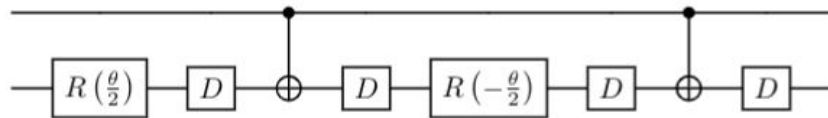
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[4] A. Barenco et al., 1995.

Assume that D is self-inverse and R is a rotation satisfying $(DXD)R(\theta)(DXD) = R(-\theta)$ and $R(\theta)R(-\theta) = I$.

The following is a decomposition of $C(R(\theta))$:

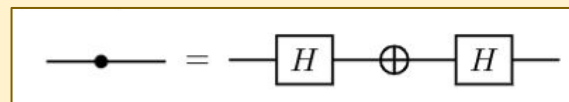
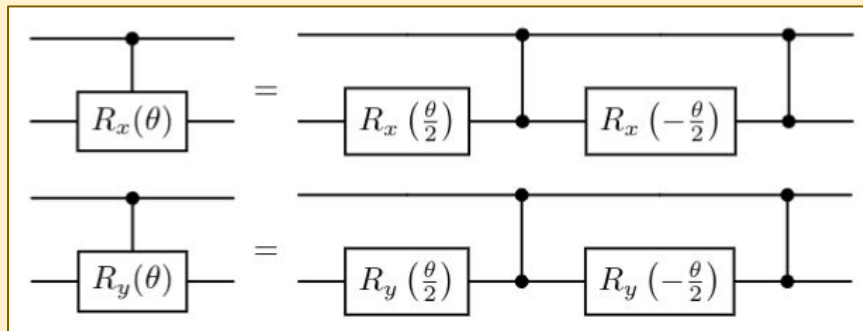


Wider Classes of Gate Decompositions

Phase

Controlled

Assume that $U = CXBXA$ and $CBA = I$.



$= I$.

Rotations

Rotations



Questions?

Want to learn more? Contact: scott.wesley@dal.ca

Want to try LinguaQuanta? Visit: github.com/onestruggler/qasm-guipper

A draft of the paper will be available through the IWQC website.

